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(54) **SECURE ONBOARDING OF USERS AND ORGANIZATIONS ON MOBILITY-AS-A-SERVICE (MaaS) PLATFORM**

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See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,835,436 A * 11/1998 Ooishi G11C 11/4096 365/230.03
5,890,194 A * 3/1999 Kim G11C 8/18 711/209
11,488,159 B1 * 11/2022 Murao G06Q 20/0855
2023/0036353 A1 * 2/2023 Wakabayashi H04M 15/61

OTHER PUBLICATIONS

Stockburger, et al., "Blockchain-enabled decentralized identity management: The case of self-sovereign identity in public transportation", Blockchain: Research and Applications, Elsevier, ScienceDirect, vol. 2, Issue 2, Jun. 2021, 18 pages.

(Continued)

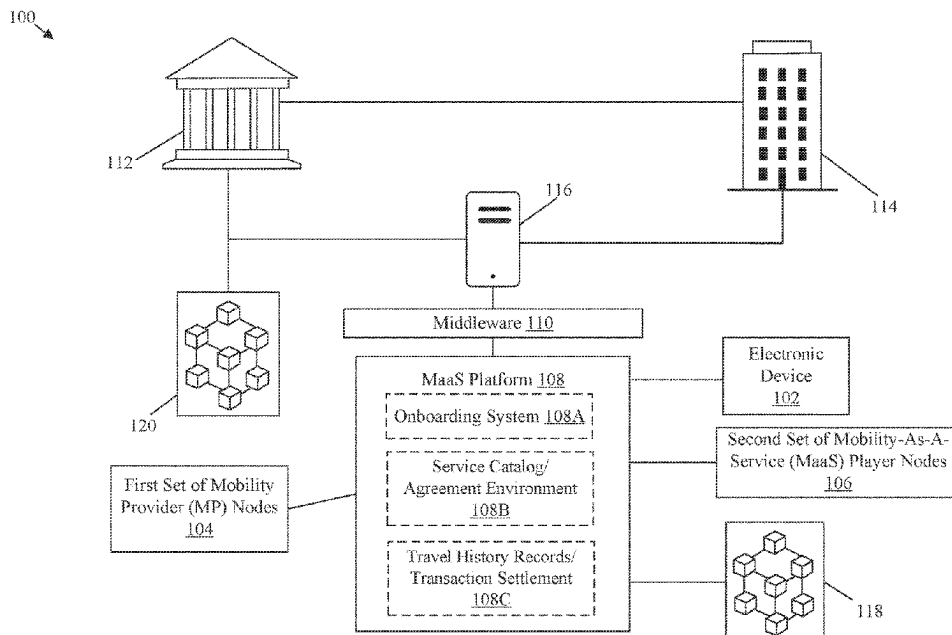
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(57) **ABSTRACT**

An electronic device for secure onboarding of users and organizations on a mobility-as-a-service (MaaS) platform is disclosed. A set of onboarding agreements is received from nodes of a MaaS network. A set of service and agreement templates are generated to create a MaaS node group corresponding to the set of nodes. The set of onboarding agreements and the set of service and agreement templates are transmitted to the MaaS node group. An onboarding request is received from a first node of the MaaS network. A decentralized identity (DID) of the first node is verified. A set of parameters and a service agreement associated with the first node is transmitted to the first node. The service agreement is transmitted to the MaaS node group. An Application Programming Interface (API) key of a data hub is generated for the MaaS network to onboard the first node to the MaaS node group.

20 Claims, 12 Drawing Sheets



(56)

References Cited

OTHER PUBLICATIONS

Bothos, et al., “Leveraging Blockchain for Open Mobility-as-a-Service Ecosystems”, WI '19 Companion: IEEE/WIC/ACM International Conference on Web Intelligence—Companion vol. Oct. 14, 2019, pp. 292-296.

“Self-Sovereign Identity meets Mobility-as-a-Service”, MyEGO, Wunder Mobility summit, Oct. 11, 2021, 03 pages.

* cited by examiner

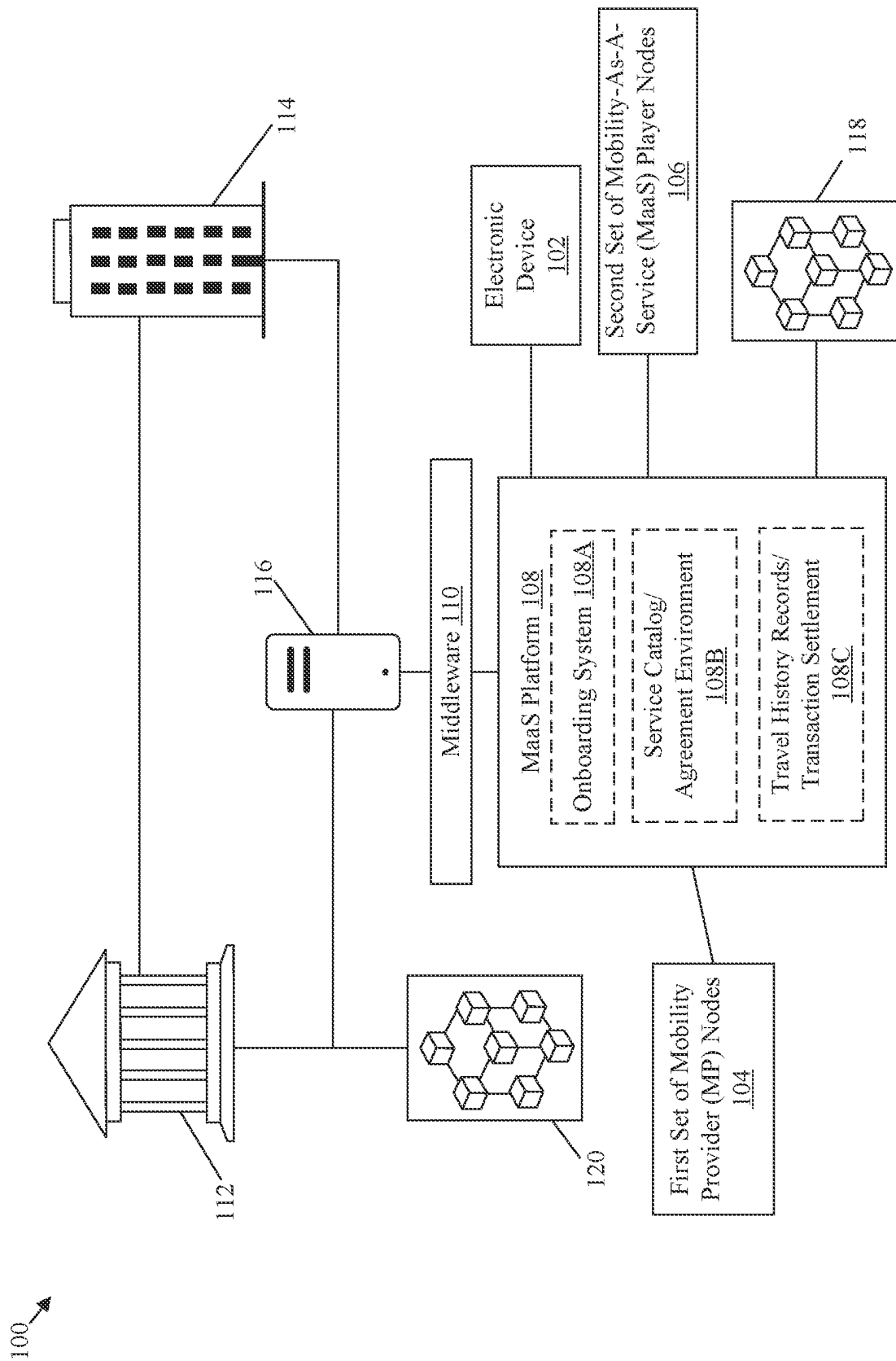


FIG. 1

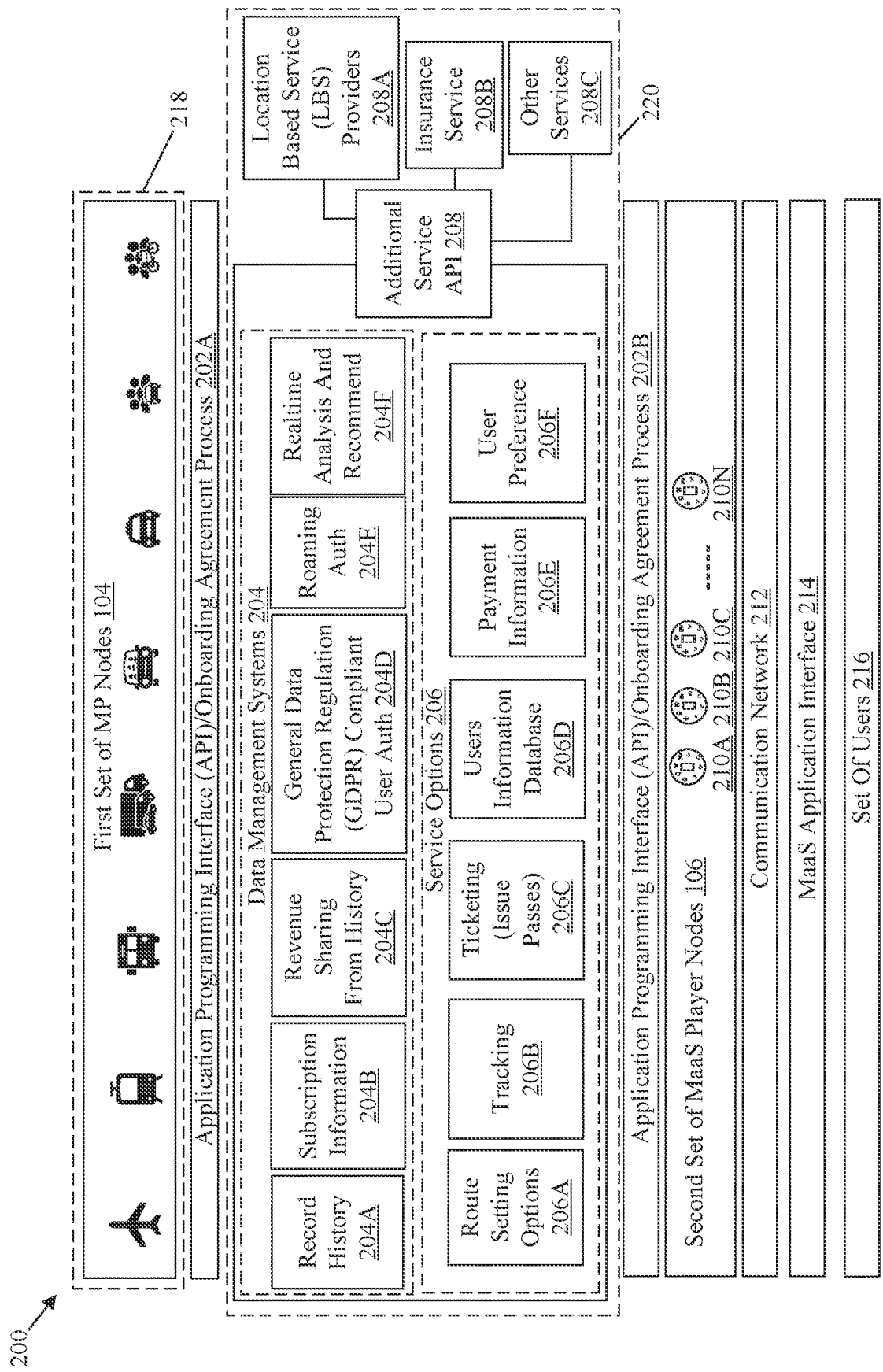


FIG. 2

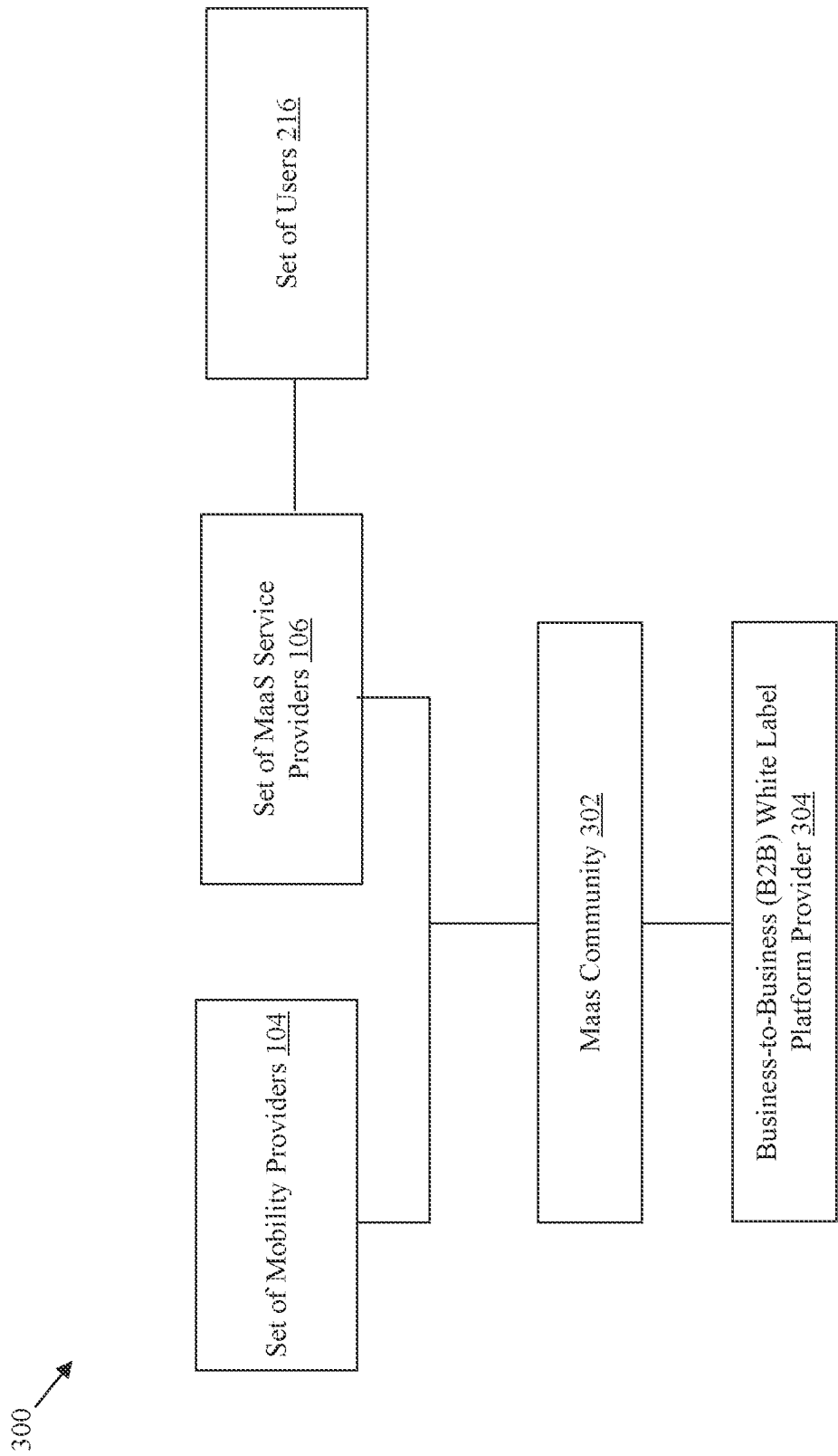


FIG. 3

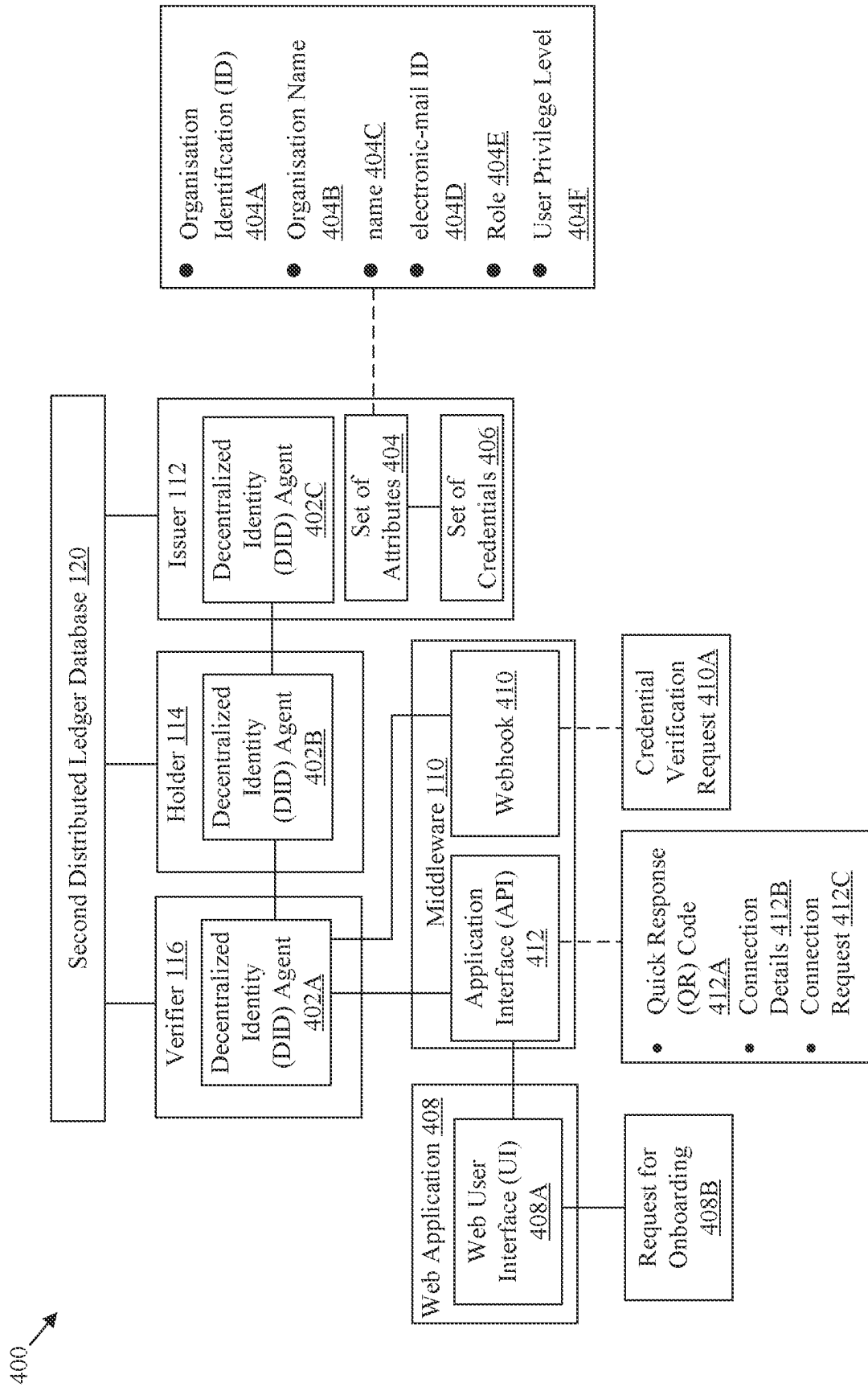


FIG. 4A

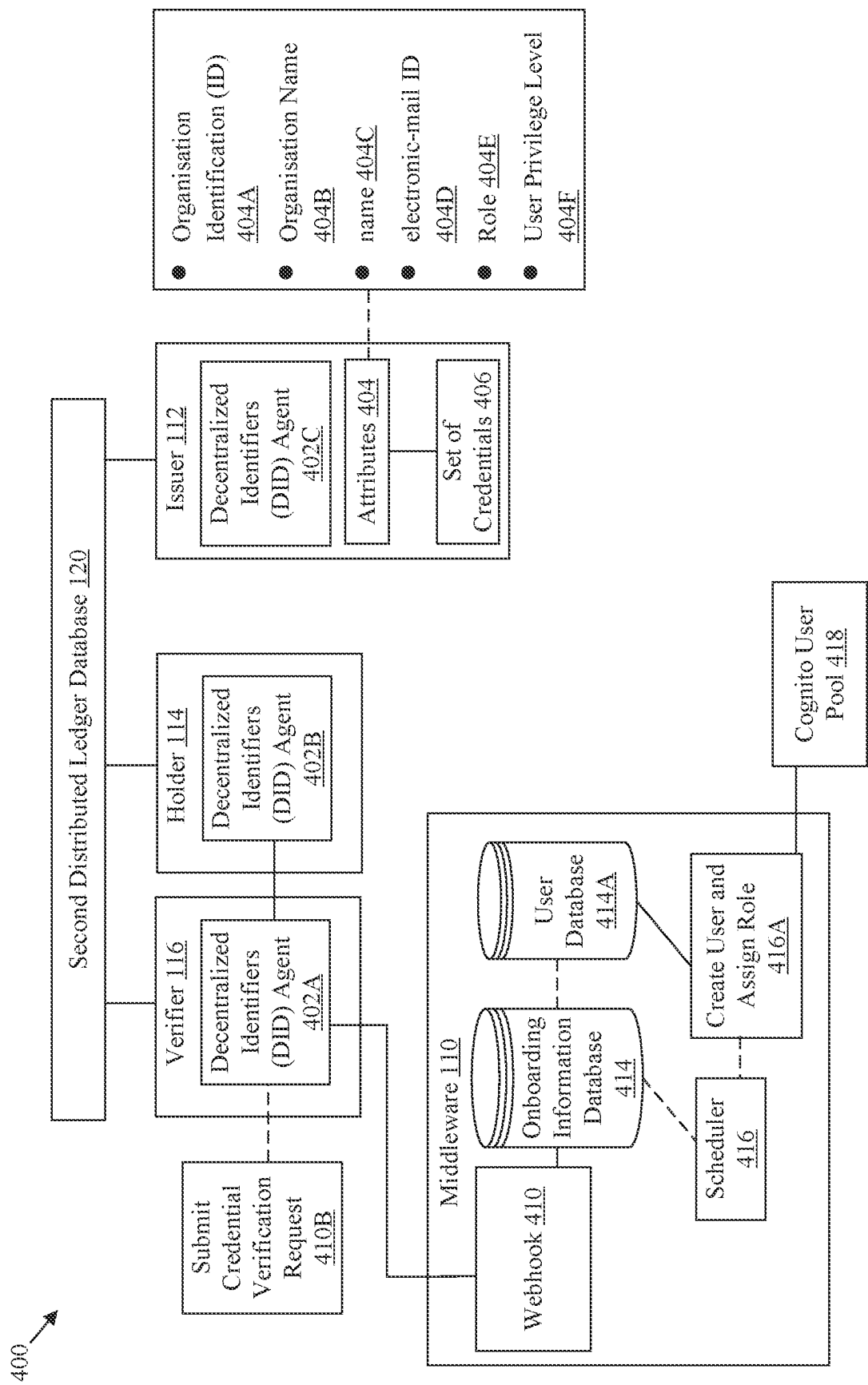


FIG. 4B

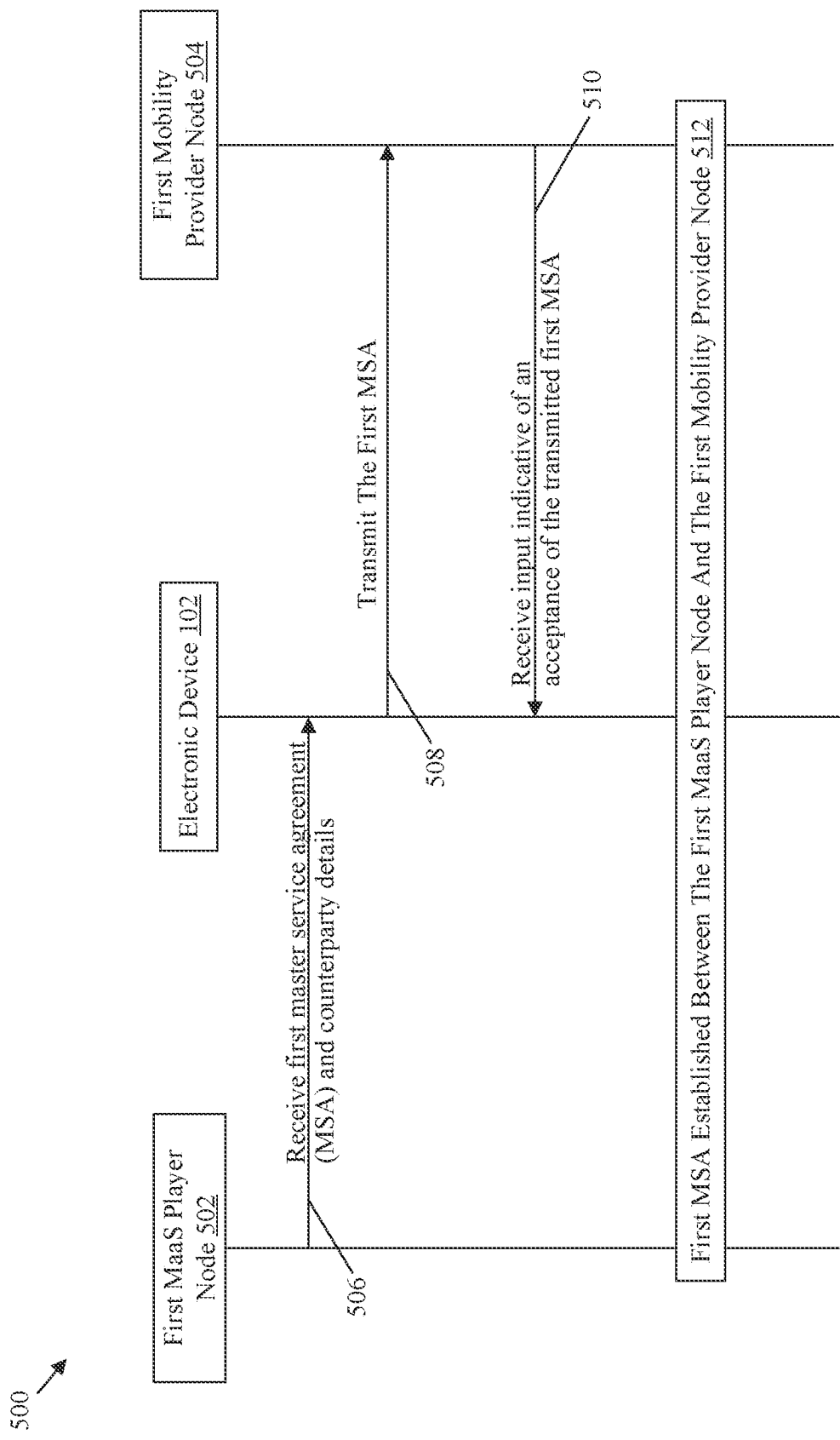


FIG. 5

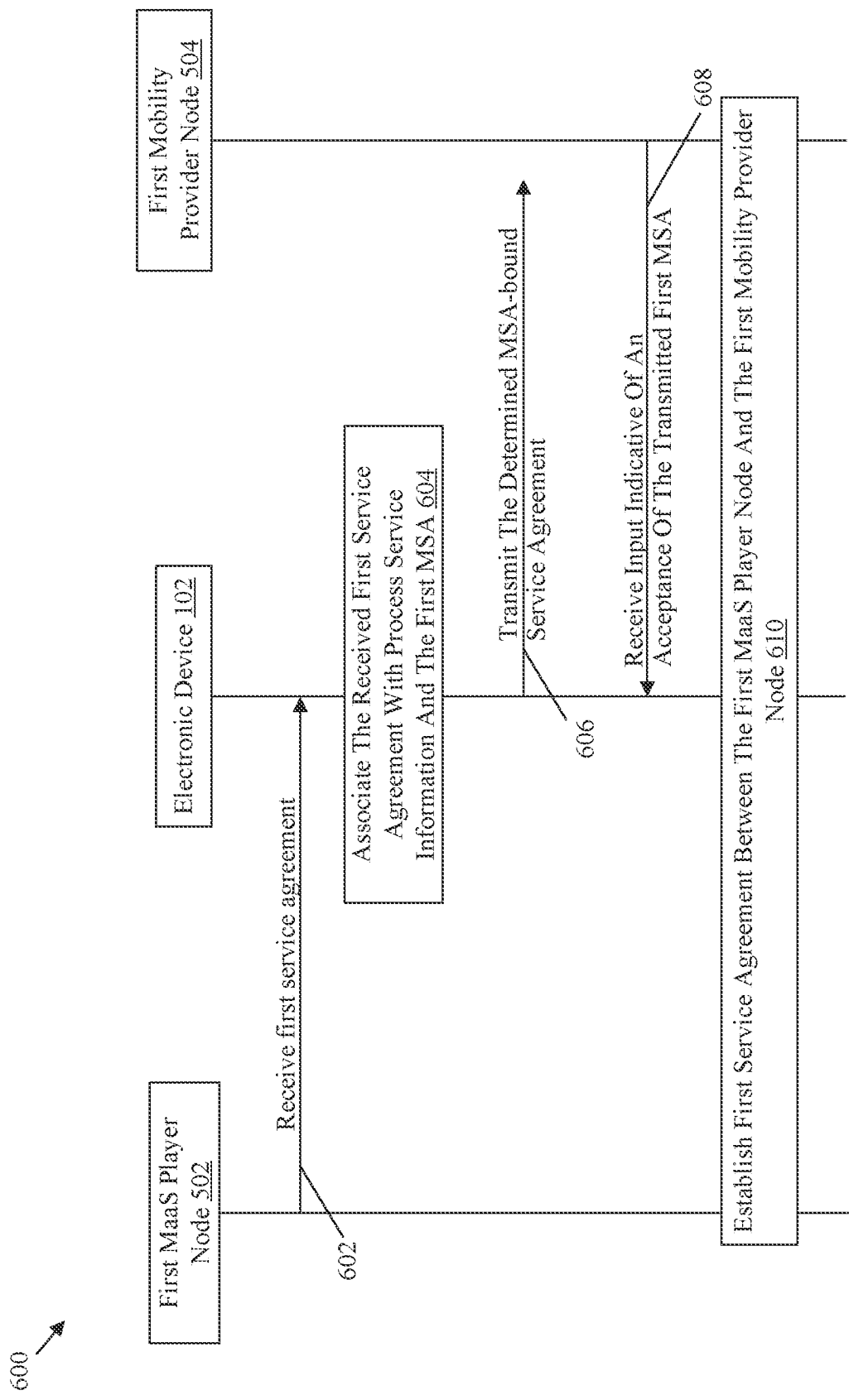


FIG. 6

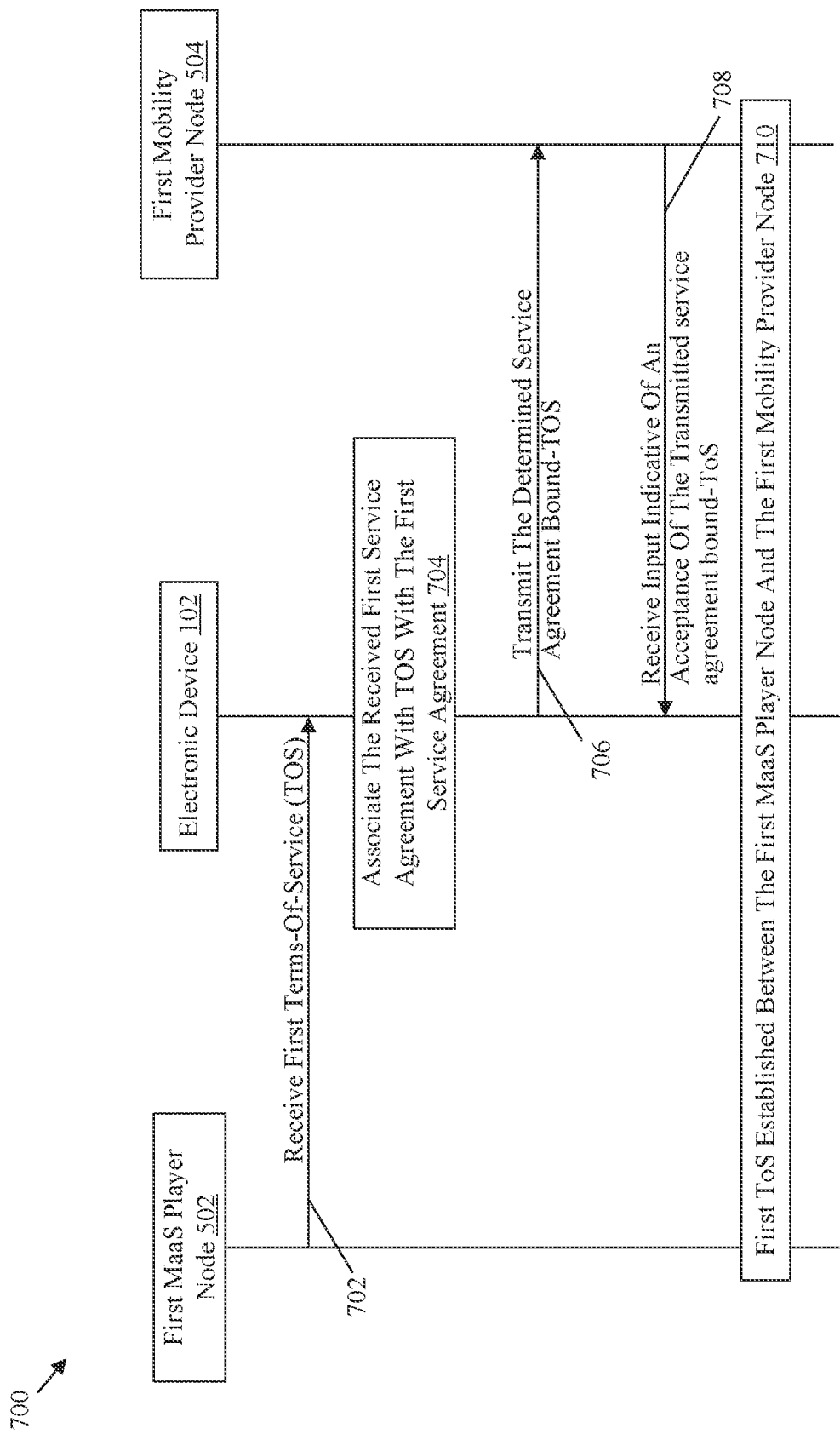


FIG. 7

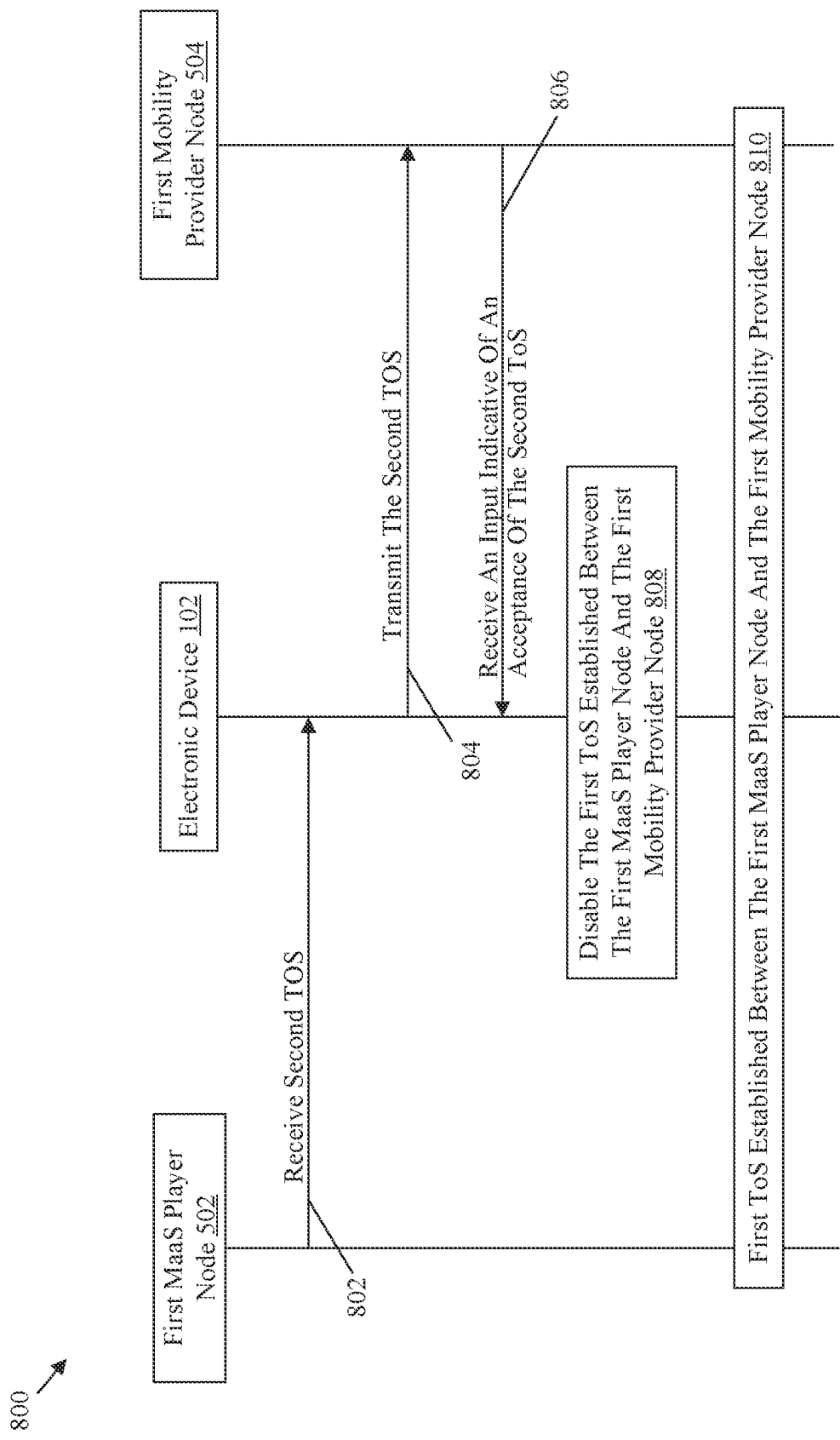


FIG. 8

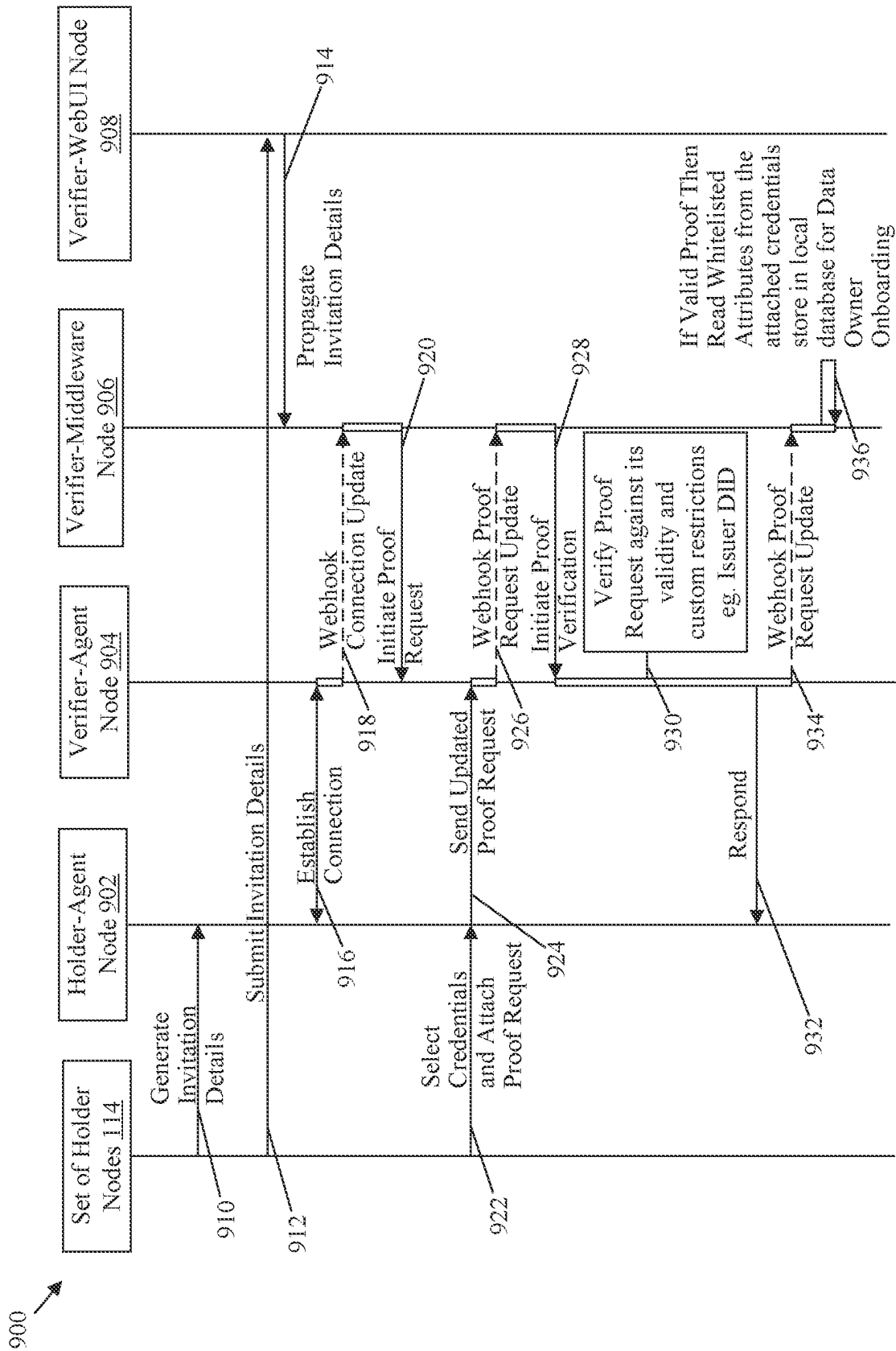


FIG. 9

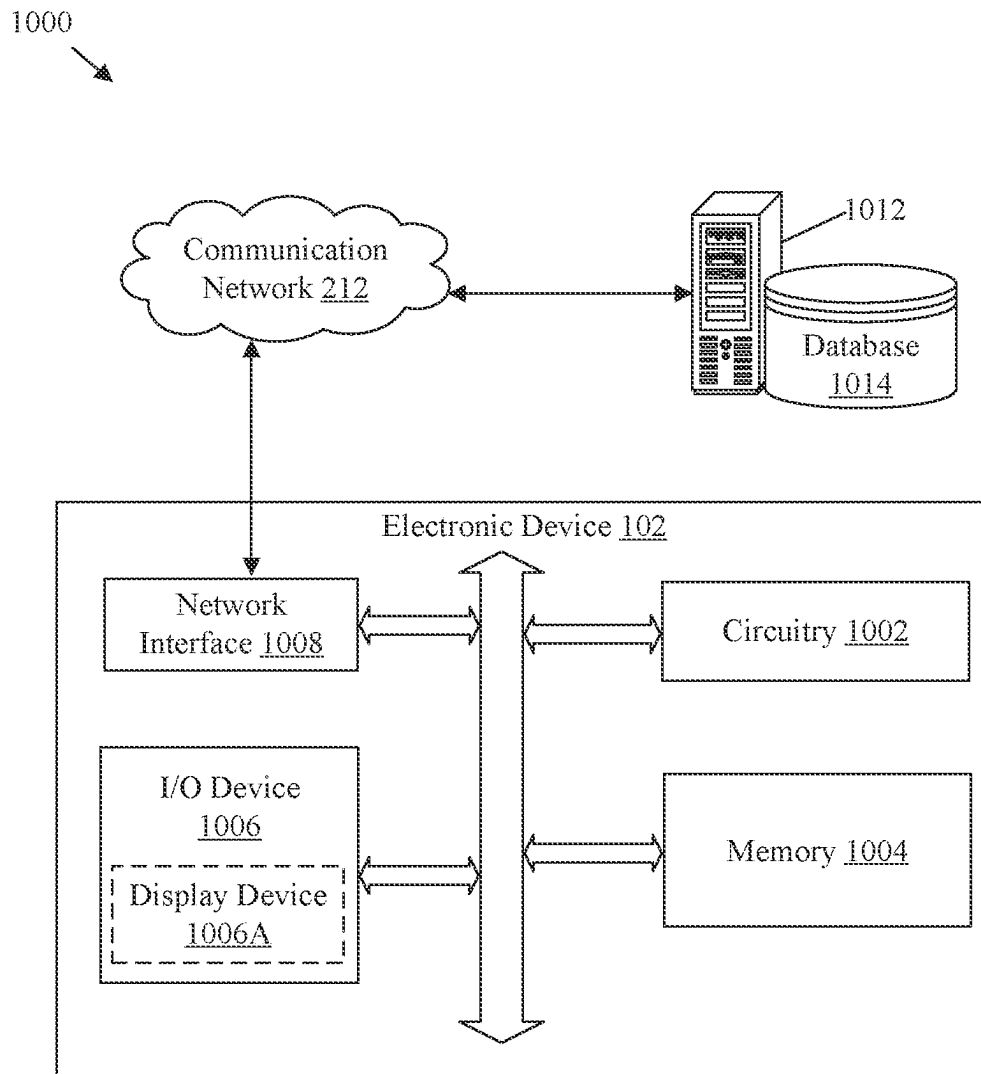


FIG. 10

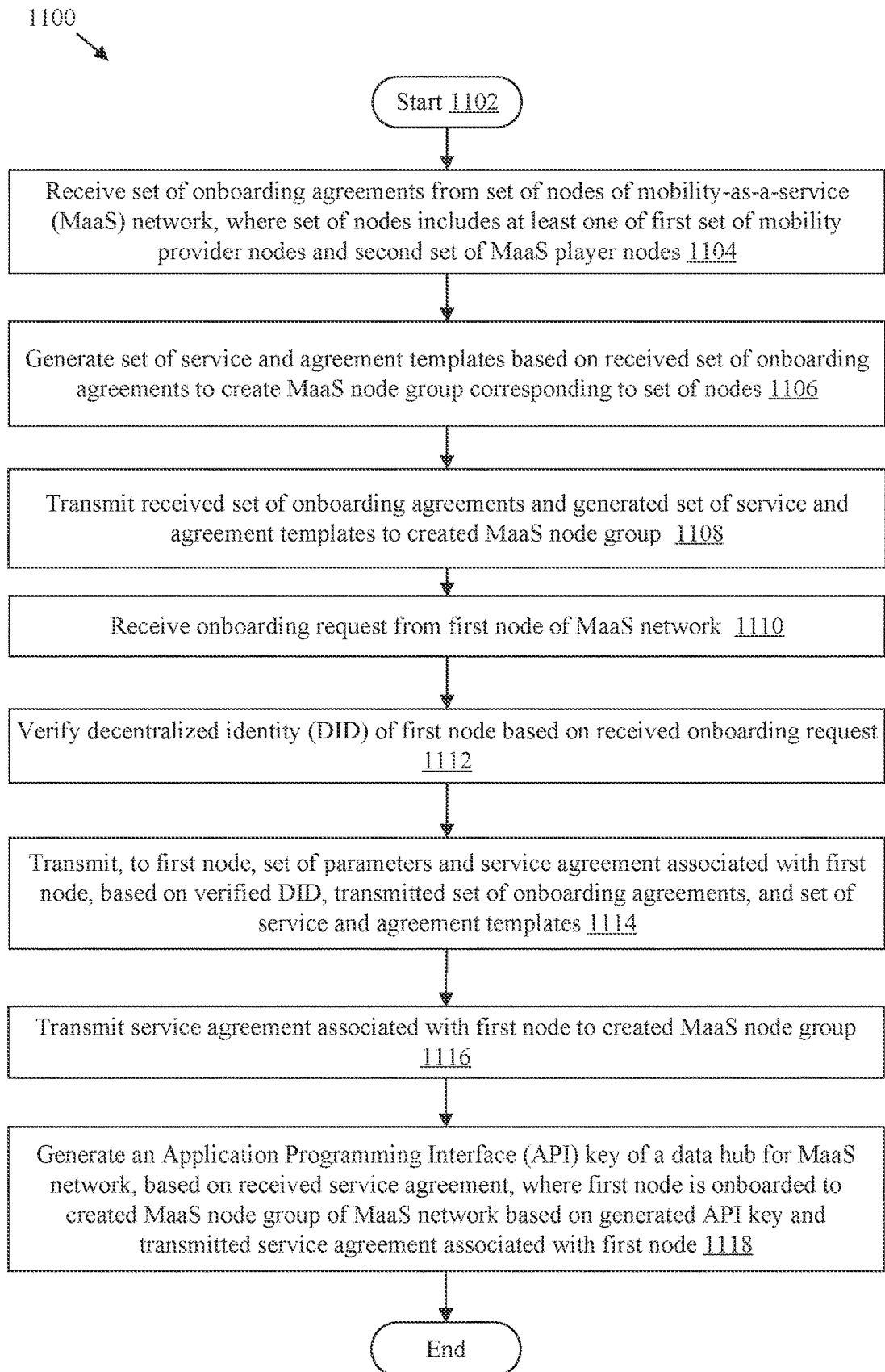


FIG. 11

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SECURE ONBOARDING OF USERS AND ORGANIZATIONS ON MOBILITY-AS-A-SERVICE (MaaS) PLATFORM

CROSS-REFERENCE TO RELATED APPLICATIONS/INCORPORATION BY REFERENCE

None

FIELD

Various embodiments of the disclosure relate to mobility-as-a-service (MaaS) platform. More specifically, various embodiments of the disclosure relate to an electronic device and method for secure onboarding of users and organizations on mobility-as-a-service (MaaS) platform.

BACKGROUND

Onboarding refers to a process in which a new user or a new organization is registered with an online portal (e.g., mobility-as-a-service (MaaS) platform) to offer a service or consume a service, through the online portal. The process of onboarding may include activities, such as, a setup of credentials, configuration of email addresses, creation of roles, and assignment of privileges etc., for users (such as, personnel of an organization). The onboarding of a user to the online portal may enable the user to access a platform or services associated with the online portal. For example, a MaaS platform may be a digital platform that may be used to manage travel-related services. The MaaS platform may provide end-to-end travel services that may include various transportation modes such as public transport modes (such as train, airplane, or bus), commercial transport services (such as bike and car sharing and taxis) through a single digital platform. The MaaS platform may also provide payment services and transaction processing services for each journey (which may include multiple legs) managed by the MaaS platform.

The MaaS platform may require a coordinated interaction between various entities (such as, multiple organizations or users) to provide the end-to-end travel services. For example, the various entities may provide transportation services, transaction processing services, payment services, and administrative (or logistic) services to enable the MaaS platform to operate and provide the end-to-end travel services to riders. Typically, for such a MaaS platform, there may be no streamlined process to onboard the various entities, such as, users, organizations, and personnel associated with organizations. The onboarding of a new user or a new organization to the MaaS platform may be a cumbersome and a time-consuming process that may require manual verification or authorization. For example, manual verification of new users through email may be required. In such case, the onboarding process may be an ad-hoc process. During the onboarding process, sensitive information (such as, name, email address, or organization name etc.) associated with the user, the organization, or the personnel associated with the organization may be available to third party administrators of the MaaS platform. The availability of such sensitive information to the third parties may give rise to confidentiality and privacy concerns.

Limitations and disadvantages of conventional and traditional approaches will become apparent to one of skill in the art, through comparison of described systems with some

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aspects of the present disclosure, as set forth in the remainder of the present application and with reference to the drawings.

SUMMARY

An electronic device and method for secure onboarding of users and organizations on mobility-as-a-service (MaaS) platform is provided substantially as shown in, and/or described in connection with, at least one of the figures, as set forth more completely in the claims.

These and other features and advantages of the present disclosure may be appreciated from a review of the following detailed description of the present disclosure, along with the accompanying figures in which like reference numerals refer to like parts throughout.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram that illustrates an exemplary network environment for secure onboarding of users and organizations on a mobility-as-a-service (MaaS) platform, in accordance with an embodiment of the disclosure.

FIG. 2 is a block diagram that illustrates an exemplary layered architecture diagram of a mobility-as-a-service platform for secure onboarding of users and organizations on the MaaS platform, in accordance with an embodiment of the disclosure.

FIG. 3 is an exemplary block diagram that illustrates an onboarding agreement system, in accordance with an embodiment of the disclosure.

FIGS. 4A and 4B illustrate exemplary architecture for credential request and verification of a node to be onboarded to a MaaS network, in accordance with an embodiment of the disclosure.

FIG. 5 is an exemplary sequence diagram that illustrates establishment of a master service agreement (MSA) between nodes of a Mobility-as-a-Service (MaaS) platform, in accordance with an embodiment of the disclosure.

FIG. 6 is an exemplary sequence diagram that illustrates establishment of service agreement between nodes of a Mobility-as-a-Service (MaaS) platform, in accordance with an embodiment of the disclosure.

FIG. 7 is an exemplary sequence diagram that illustrates establishment of a first terms of service (ToS) between nodes of a Mobility-as-a-Service (MaaS) platform, in accordance with an embodiment of the disclosure.

FIG. 8 is an exemplary sequence diagram that illustrates establishment of a second terms of service (ToS) between nodes of a Mobility-as-a-Service (MaaS) platform to override a previous version of ToS established between the nodes, in accordance with an embodiment of the disclosure.

FIG. 9 is an exemplary sequence diagram that illustrates secure onboarding of users and organizations on a mobility-as-a-service (MaaS) platform, in accordance with an embodiment of the disclosure.

FIG. 10 is a block diagram that illustrates an exemplary electronic device of FIG. 1, in accordance with an embodiment of the disclosure.

FIG. 11 is a flowchart that illustrates operations of an exemplary method for secure onboarding of users and organizations on a mobility-as-a-service (MaaS) platform, in accordance with an embodiment of the disclosure.

DETAILED DESCRIPTION

The following described implementations may be found in the disclosed electronic device and method for secure

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onboarding of users and organizations on mobility-as-a-service (MaaS) platform. The disclosed electronic device may be a part of a federated transportation management system that may facilitate multiple homogeneous or heterogeneous mobility providers and their infrastructure, such as ticketing gates, applications, and/or Point of Sale (POS) devices to operate on the MaaS network to provide various mobility services. Each mobility provider may have secure data ownership and may control co-use of relevant transaction records through a distributed ledger. This may enhance connectivity between the various mobility providers. The electronic device may further enhance roaming management, and contract management between the different mobility providers, based on the controlled co-use of the relevant transaction records through the distributed ledger. Each mobility provider in the MaaS network may be associated with the MaaS network through a service agreement with certain terms or service and a related privacy policy. Before a node of a mobility provider joins the MaaS network, the mobility provider node may be on-boarded to the MaaS network, based on the service agreement, terms of service, and the privacy policy. On acceptance of the service agreement and based on the terms of service and the privacy policy, the node of the mobility provider may be on-boarded and admitted to the MaaS network.

Exemplary aspects of the disclosure provide an electronic device for secure onboarding of users and organizations on mobility-as-a-service (MaaS) platform. The electronic may receive a set of onboarding agreements from a set of nodes of a mobility-as-a-service (MaaS) network. The set of nodes may include at least one of a first set of mobility provider nodes and a second set of MaaS player nodes. The electronic device may generate a set of service and agreement templates based on the received set of onboarding agreements to create a MaaS node group corresponding to the set of nodes. The electronic device may transmit the received set of onboarding agreements and the generated set of service and agreement templates to the created MaaS node group. The electronic device may receive an onboarding request from a first node of the MaaS network. Based on the received onboarding request, the electronic device may verify a decentralized identity (DID) of the first node. Based on the verified DID, transmitted set of onboarding agreements, and the set of service and agreement templates, the electronic device may transmit, to the first node, a set of parameters and a service agreement associated with the first node. Based on the received service agreement, the electronic device may generate an Application Programming Interface (API) key of a data hub for the MaaS network. The first node may be onboarded to the created MaaS node group of the MaaS network based on the generated API key and the transmitted service agreement associated with the first node.

Typically, for a MaaS platform, there may be no streamlined process to onboard the various entities, such as, users, organizations, and personnel associated with organizations. The onboarding of a new user or a new organization to the MaaS platform may be a cumbersome and a time-consuming process that may require manual verification or authorization. For example, manual verification of new users through email may be required. In such case, the onboarding process may be an ad-hoc process. During the onboarding process, sensitive information (such as, name, email address, or organization name etc.) associated with the user, the organization, or the personnel associated with the organization may be available to third party administrators of the MaaS

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platform. The availability of such sensitive information to the third parties may give rise to confidentiality and privacy concerns.

The disclosed electronic device may receive an onboarding request from a first node (such as the first node) of the MaaS network (such as the MaaS platform). Based on the received onboarding request, a decentralized identity (DID) (such as the DID) of the first node may be verified. Further, based on the verified DID, transmitted set of onboarding agreements, and the set of service and agreement templates, a set of parameters and a service agreement associated with the first node may be transmitted to the first node. Further, the electronic device may transmit the service agreement associated with the first node to the created MaaS node group. Thereafter, based on the received service agreement, an Application Programming Interface (API) key of a data hub for the MaaS network may be generated. Thus, the first node may be onboarded to the created MaaS node group of the MaaS network based on the generated API key and the transmitted service agreement associated with the first node. For example, the first node, that may correspond to any computing device of a user, an organization, or personnel associated with the organization, may be onboarded based on generation of the API key and transmission of the service agreements. Further, the API key may be used by the onboarded user (or the organization, or personnel associated with the organization) to share data (such as, revenue data, travel data, cyclic revenue, or travel data). Thus, a streamlined and secure on-boarding of new nodes to the MaaS platform may be achieved without concerns related to confidentiality and privacy of various entities (such as, the users/organizations/personnel) to be on-boarded. After an entity is successfully on-boarded to the MaaS platform, the entity may start to communicate with the other entities of the MaaS platform and offer service (for example, in case of MaaS player nodes or mobility provider nodes) or consume services (for example, in case of end-users or mobility provider nodes) associated with the MaaS platform.

FIG. 1 is a block diagram that illustrates an exemplary network environment for secure onboarding of users and organizations on a mobility-as-a-service (MaaS) platform, in accordance with an embodiment of the disclosure. With reference to FIG. 1, there is shown a block diagram of a network environment **100**. The network environment **100** (such as mobility-as-a-service (MaaS) network **100**) may include an electronic device **102**, a first set of mobility provider (MP) nodes **104**, a second set of MaaS player nodes **106**, a mobility-as-a-service (MaaS) platform **108**, a first distributed ledger database **118**, and a middleware **110** (which may be layered software that interfaces the various components of the MaaS network **100**). The MaaS network **100** may include a network of nodes, such as an issuer node **112**, a set of holder nodes **114**, a verifier node **116** and a second distributed ledger database **120**. In an embodiment, the MaaS platform **108** may include an onboarding system **108A**, a service catalog/agreement environment **108B**, and a system for travel history records/transaction settlement **108C**. The issuer node **112** may be communicatively coupled to the set of holder nodes **114**, the verifier node **116** and the second distributed ledger database **120**. The set of holder nodes **114** may be communicatively coupled to the verifier node **116** and the issuer node **112**. The verifier node **116** may be communicatively coupled to the issuer node **112**, the set of holder nodes **114**, the second distributed ledger database **120**, and the middleware **110**. The electronic device **102**, the first set of mobility provider nodes **104**, the second set of MaaS player nodes **106**, the first distributed ledger

database **118**, and the middleware **110** may be communicatively coupled to the MaaS platform **108**.

The electronic device **102** may include suitable logic, circuitry, interfaces, and/or code that may be configured to receive a set of onboarding agreements (for example, a master service agreement, a terms-of-service agreement, or a privacy policy agreement) from a set of nodes of a mobility-as-a-service (MaaS) network (e.g., the MaaS network **100**). The MaaS network **100** may include at least one of the issuer node **112** associated with a credential issuing authority, the set of holder nodes **114** corresponding to the set of nodes, the verifier node **116** associated with a MaaS network owner, and the second distributed ledger database **120**. The set of nodes may include at least one of the first set of MP nodes **104** and the second set of MaaS player nodes **106**. Based on the received set of onboarding agreements, the electronic device **102** may create a MaaS node group (for example, the first set of MP nodes **104** or the second set of MaaS player nodes **106**) that may correspond to the set of nodes. Further, the electronic device **102** may generate a set of service and agreement templates. Further, the electronic device **102** may transmit the received set of onboarding agreements and the generated set of service and agreement templates to the created MaaS node group. The electronic device **102** may receive an onboarding request from a first node (for example, a user, an organization, or personnel associated with the organization) of the MaaS network **100**. Based on the received onboarding request, the electronic device **102** may verify a decentralized identity (DID) of the first node. Based on the verified DID, the transmitted set of onboarding agreements, and the set of service and agreement templates, the electronic device **102** may transmit a set of parameters and a service agreement associated with the first node to the first node. The DID of a node (e.g., the first node) may correspond to a unique identifier that may be used to verify a digital identity of the node using a decentralized approach. For example, the DID may include a unique machine ID of the node and a global or a local timestamp associated with the node. In an embodiment, the verification of an entity (e.g., the first node) of the MaaS network **100** through a DID of the entity may enable the entity to have a greater control over sensitive identification information of the entity. Based on the DID of the entity, the entity may be authenticated using a decentralized application of cryptographic techniques on the digital identity of the entity.

The electronic device **102** may transmit the service agreement associated with the first node to the created MaaS node group. Further, the electronic device **102** may generate an Application Programming Interface (API) key of a data hub for the MaaS network **100**, based on the received service agreement. Thereafter, the first node may be onboarded to the created MaaS node group of the MaaS network **100** based on the generated API key and the transmitted service agreement associated with the first node.

The first set of MP nodes **104** may include suitable logic, circuitry, code, and/or interfaces that may be configured to store transaction records associated with a respective mobility provider. For example, the first set of MP nodes **104** may store transaction records associated with a first mobility provider. In an embodiment, each of the transaction records stored on each MP node may be associated with a two-party transaction. For example, the transaction records stored on an MP node of a certain mobility provider may be associated with transactions that may involve the mobility provider and a certain MaaS provider. The transaction records may include records of trips of users. Each trip may correspond to a mobility service that may be provided by the mobility

provider in at least one leg of a journey of a rider who uses the mobility service. The first set of MP nodes **104** may be referred to as nodes of a distributed ledger (such as, the first distributed ledger database **118**) that may store transaction records of the various mobility providers of the MaaS network **100**. In an embodiment, each MP node may be implemented as, for example, one of an edge device, an edge node, or a distributed ledger node with a fog or cloud computing capability.

In accordance with an embodiment, each of the first set of MP nodes **104** may be associated with a separate mobility provider of the MaaS network **100**. The MaaS mobility service may be provided by homogeneous mobility providers (such as multiple cab ride provider companies or multiple rail companies) or heterogeneous mobility providers through a homogenous group of devices, applications, or ticketing gates, or a heterogeneous group of ticketing gates, applications, and Point of Sale (PoS) devices. The MaaS mobility service may be a combination of individual service offerings of one or more homogenous or heterogeneous mobility providers. The MaaS mobility service may include, for example, a train service, a bus service, a taxi/cab service, a metro rail service, an air-plane service, a fleet service, a ride hailing service, a car sharing service, a carpool service, a car rental service, a bike sharing service, or a combination thereof.

The second set of MaaS player nodes **106** may include suitable logic, circuitry, code, and/or interfaces that may be configured to prepare a transportation route, that may include one or more mobility providers of the first set of MP nodes **104** based on a demand of a customer, such as, a rider who consumes a mobility service. The MaaS player may provide personalized services to customers on a single platform. The MaaS provider thus may have a function to aggregate transport services from many transport service operators (such as the mobility providers). The MaaS players may comprise any of commercial MaaS player, public transport provider, and a government organization. The MaaS players may further manage data associated with the users that may include storage of the data and to perform analytics on the data for optimization. The MaaS player of the second set of MaaS player nodes **106** may be categorized into two categories, based on role, that may include a MaaS integrator and a MaaS operator. The MaaS integrator may correspond to the role where the MaaS players assemble the transport services from multiple transport service providers that may include activities such as technical integration, contract management and financial clearing. The MaaS operator may correspond to the role where the MaaS player provides a MaaS service to end-users or customers through a single platform.

The MaaS platform **108** may include suitable logic, circuitry, code, and/or interfaces that may be configured to provide support for trip planning, ticketing, and payment services across multiple transportation providers (such as the mobility providers). The MaaS platform **108** may support mobile and web applications that may be accessed through user devices associated with users or customers. The MaaS platform **108** may collect, and store data associated with the users. The data associated the user may include, travel history, travel time, frequency of travel, and the like. Further, the MaaS platform **108** may analyze the data to optimize travel networks and calibrate supply and demand. The MaaS platform **108** may further include, the onboarding system **108A**, the service catalog/agreement environment **108B**, and a system for the travel history records/transaction settlement **108C**. The onboarding system **108A** may be used to inte-

grate (such as, to onboard) or disintegrate (such as, to deboard) a node from the MaaS network **100**.

The system for travel history records/transaction settlement **108C** may be used to record and store data associated with user or nodes of the MaaS network **100**. The data associated with user or nodes may include company information associated with the first MP, service item information associated with each of the first set of transaction records, ticket sales data associated with each of the first set of transaction records, fare price master data associated with the first MP, service price master data associated with the first MP, and/or ticket transaction data associated with each of the first set of transaction records. For example, the company information associated with the first MP may include a name, an address, a number of vehicles, and/or type of vehicles, associated with a company corresponding to the first MP. The service item information associated with each of the first set of transaction records may include a date/time of a trip leg, a duration of the trip leg (based on get-in/get-out transaction messages), a distance travelled on the trip leg, and/or a type of vehicle associated with the trip leg, associated with each of the first set of transaction records. The ticket sales data associated with each of the first set of transaction records may include a price of a ticket for the trip, and/or a price of the trip leg associated with the first MP, associated with each of the first set of transaction records. The fare price master data associated with the first MP may include a trip-leg distance-based fare table, and/or a trip-leg duration-based fare table associated with the first MP. The service price master data associated with the first MP may include a traffic surcharge, a wait-time surcharge, a trip-leg duration surcharge, a day-of-week based surcharge, a time-of-day based surcharge, a holiday surcharge, and/or a service tax, associated with the first MP. The ticket transaction data associated with each of the first set of transaction records may include, but is not limited to, transaction data corresponding to a trip-leg associated with the first MP, from ticket data in each of the first set of transaction records.

The middleware **110** may include suitable logic, circuitry, code, and/or interfaces that may be configured to enable communication and message passing between various software applications, databases, application programming interfaces, and other remote or local application components. Examples of the middleware **110** may include, but are not limited to, a database middleware, an application server middleware, message-oriented middleware, transaction-processing middleware, and web middleware.

The issuer node **112** may include suitable logic, circuitry, code, and/or interfaces that may be configured to generate credentials associated with each holder node of the set of holder nodes **114**. The generated credentials may correspond to a set of attributes including at least one of an organization name, an organization ID, a name and electronic-mail ID of an administrator user, a role of a user of a node, or a user privilege level associated with a user of a node. The issuer node **112** may be associated with a government organization or a private organization. The issuer node **112** may define a credential template. The credential template may further include certain attributes, for example, ID of organization, organization name, admin username (such as the personnel associated with the organization), email address of admin, role, and responsibility of the admin. In an example, the issuer node **112** may manage a creation of digital identities of the set of holder nodes **114** and an authorization of the set of holder nodes **114** to the MaaS platform **108**.

The set of holder nodes **114** may include suitable logic, circuitry, code, and/or interfaces that may be configured to participate as nodes in the MaaS network **100** and be communicatively coupled to the MaaS platform **108**. Each of the set of holder nodes **114** may receive credential information associated with the corresponding holder node from the issuer node **112**. The received credential information may correspond to a digital identification of the holder node. The holder node may transmit an onboarding request to the electronic device **102**, to onboard the holder node to the MaaS network **100**. The onboarding request of the holder node may be authenticated by the verifier node **116** based on the credential information of holder node. The onboarding of the holder node to the MaaS network **100** may enable the holder node to join the MaaS network **100** and communicate with the MaaS platform **108** and the other holder nodes of the set of holder nodes **114**. In an example, the set of holder nodes **114** may correspond to a mobility provider node or a MaaS player node.

The verifier node **116** may include suitable logic, circuitry, code, and/or interfaces that may be configured to receive a credential verification request associated with a first node (of a set of nodes of the MaaS network **100**), based on an onboarding request associated with the first node. Further, the verifier node **116** may query a holder node (such as, a holder node associated with the same mobility provider or MaaS player as that of the first node) of the set of holder nodes **114** for verified credentials of the first node, based on the received credential verification request. Further, the verifier node **116** may receive, from the holder node, the verified credentials of the first node, based on the query. The DID of the first node may be verified based on the received verified credentials of the first node. Once the verification is completed, the first node may be onboarded. In an example, the verifier node **116** may be associated with a governing organization or administrator of the MaaS network **100**.

In an embodiment, the first distributed ledger database **118** may be configured store transaction records associated with a MaaS mobility service. The first distributed ledger database **118** may correspond to a blockchain common database associated with the MaaS mobility service. The transaction records associated with the MaaS mobility service may be included in a set of state objects, such as an initial state object and an updated version of the initial state object. In an embodiment, the second distributed ledger database **120** may be configured to store credential information associated with nodes of the MaaS network **100**. The stored credential information may be used for decentralized identification (DID) based verification or authentication of the nodes. The second distributed ledger database **120** may correspond to a decentralized identification (DID) blockchain database associated with the MaaS network **100**. Each state object and/or credential information may correspond to a smart contract, a contract code (or rules of a transaction upon which parties to the transaction agree), and state properties (that may be updated when the transaction records and/or credential information may be updated based on transaction messages from the publisher node and/or credential creation by an issuer node). By way of example, and not limitation, each of the first distributed ledger database **118** and the second distributed ledger database **120** may be a Corda blockchain, an Ethereum blockchain, or a Hyperledger blockchain. For example, U.S. patent application Ser. No. 17/178,185 filed on Feb. 17, 2021, discloses details related to a distributed ledger database (such as, the first distributed ledger database **118** and the second distributed ledger database **120**).

In operation, the electronic device **102** may be configured to receive a set of onboarding agreements from a set of nodes of a mobility-as-a-service (MaaS) network, such as, the MaaS network **100**. The set of nodes may include at least one of a first set of mobility provider nodes (e.g., the first set of MP nodes **104**) and a second set of MaaS player nodes (e.g., the second set of MaaS player nodes **106**). The electronic device **102** may generate a set of service and agreement templates based on the received set of onboarding agreements to create a MaaS node group corresponding to the set of nodes. The electronic device **102** may transmit the received set of onboarding agreements and the generated set of service and agreement templates to the created MaaS node group. The electronic device **102** may receive an onboarding request from a first node of the MaaS network **100**. Based on the received onboarding request, the electronic device **102** may verify a decentralized identity (DID) of the first node. Based on the verified DID, the transmitted set of onboarding agreements, and the set of service and agreement templates; the electronic device **102** may transmit, to the first node, a set of parameters and a service agreement associated with the first node. Based on the received service agreement, the electronic device **102** may generate an Application Programming Interface (API) key of a data hub for the MaaS network **100**. The first node may be onboarded to the created MaaS node group of the MaaS network **100** based on the generated API key and the transmitted service agreement associated with the first node. The onboarding of the first node to the MaaS network is described further, for example, in FIGS. **2**, **3**, **4A**, **4B**, **5**, **6**, **7**, **8**, and **9**.

Typically, there may be no streamlined processes to onboard users, organizations, and personnel associated with organizations, and the on-boarding may be an ad-hoc process. As such, the onboarding of a new user or a new organization to the MaaS network **100** may be a cumbersome and a time-consuming process that may require manual verification or authorization. For example, traditional on-boarding processes may require, manual authentication via emails or in person/physical verification. It may be appreciated that the MaaS platform **108** of the present disclosure may provide a streamlined and secure process to achieve onboarding of the user or the organization on the MaaS platform **108** or the MaaS network **100**. The service catalog/agreement environment **108B** may store a set of service and agreement templates. In an embodiment, a node may generate contracts or agreements, such as, a master service agreement (MSA) contract, terms of service (ToS) agreement, or a privacy policy (PP) agreement based on the storage of the set of service and agreement templates. For example, the generated set of service and agreement templates may be skeletal formats that may be generated for certain service agreements etc. For example, if two organizations want to establish a revenue share service, they may have to establish a service agreement between them for revenue sharing.

FIG. **2** is a block diagram that illustrates an exemplary layered architecture diagram of a mobility-as-a-service platform for secure onboarding of users and organizations on the MaaS platform, in accordance with an embodiment of the disclosure. FIG. **2** is explained in conjunction with elements from FIG. **1**. With reference to FIG. **2**, there is shown a layered architecture diagram **200**. The layered architecture diagram **200** may include an application program interface (API)/onboarding agreement process **202A**, an API/onboarding agreement process **202B**, a data management system **204**, a service options **206**, an additional service API

208, a communication network **212**, a MaaS application interface **214** (or MaaS API), and a set of users **216**. The layered architecture diagram **200** may further include a physical operation layer **218** (including, for example, the first set of MP nodes **104**), a MaaS layer **220**, and a MaaS operators' layer **210**, including MaaS operators **210A**, **210B**, . . . **210N** (for example, the second set of MaaS player nodes **106**).

The data management system **204** may be a layer associated with MaaS common data and business rules. The data management system **204** may include, for example, a record history **204A**, information related to a subscription **204B**, a system for revenue sharing from history **204C**, a system for General Data Protection Regulation (GDPR) compliant user auth (authentication) **204D**, a system for roaming auth (authentication) **204E**, and a system for real-time analysis and recommendation **204F**. The service options **206** may include, for example, a route setting option **206A**, a system for tracking **206B**, a system for ticketing **206C** (for issue of tickets and passes), a user information database **206D**, payment information **206E**, and information related to user preference **206F**. The additional service API **208** may include, for example, location-based service (LBS) providers **208A**, an insurance service **208B**, and other services **208C**.

The layered architecture diagram **200** may correspond to the MaaS network **100**, which may include a network of nodes, such as, a set of nodes, for example, the first set of MP nodes **104** (i.e., the physical operation layer **218**) and the second set of MaaS player nodes **106** (i.e., the MaaS operators' layer **210**). The MaaS network **100** may handle transactions (such as, transaction messages) for a MaaS mobility service associated with multiple mobility providers. In accordance with an embodiment, each of the first set of MP nodes **104** may be associated with a separate mobility provider of the MaaS network **100**. The MaaS mobility service may be provided by homogeneous mobility providers (such as multiple cab ride provider companies or multiple rail companies) or heterogeneous mobility providers through a homogenous group of devices, applications, or ticketing gates, or a heterogeneous group of ticketing gates, applications, and Point of Sale (POS) devices. The MaaS mobility service may be a combination of individual service offerings of one or more homogenous or heterogeneous mobility providers. The MaaS mobility service may include, for example, a train service, a bus service, a taxi/cab service, a metro rail service, an air-plane service, a fleet service, a ride hailing service, a car sharing service, a carpool service, a car rental service, a bike sharing service, or a combination thereof.

The physical operation layer **218** and the MaaS operators' layer **210** may interface and communicate with a MaaS platform (associated with the MaaS layer **220**), via the API/onboarding agreement process **202A** and the API/onboarding agreement process **202B**, respectively. An API may be an interface that may expose certain services associated with predefined functions of a software component to other software packages or components. For example, the first set of MP nodes **104** (such as, a node for a train service, a bus service, a taxi/cab service, a metro rail service etc.) may be integrated or interfaced with the MaaS platform **108** (i.e., the MaaS layer **220**), via the API/onboarding agreement process **202A**. In another example, the second set of MaaS player nodes **106** (such as, a node for a first MaaS operator) may be integrated or interfaced with the MaaS platform **108** (i.e., the MaaS layer **220**), via the API/onboarding agreement process **202B**. The first set of MP

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nodes **104** and the second set of MaaS player nodes **106** may be onboarded with the MaaS platform **108** based on communication of the respective node with the MaaS platform **108**, via the API/onboarding agreement process **202A** and the API/onboarding agreement process **202B**, respectively.

As shown in FIG. 2, the layered architecture diagram **200** may further include the data management system **204**. The data management system **204** may be used for implementation of a data management policy function. The data management policy function may be based on a region or area of operation of a first MP, a first MaaS player, and/or a first MaaS network as a whole. The data management policy function may also be dependent on one or more legal requirements of the region or area, which may change from time-to-time. In addition, data retention and processing rules may be governed by fine-grained parameters in a service agreement. Examples of the fine-grained parameters may include data retention period, data analytics policy, a policy for data disclosure to third parties, treatment of personally identifiable information (PII), data hashing, and the like.

The data management system **204** may be used for implementation of policies associated with any of the record history **204A**, the information related to the subscription **204B**, the system for revenue sharing from history **204C**, the system for GDPR compliant user auth **204D**, the system for roaming auth **204E**, and the system for real-time analysis and recommendation **204F**. The record history **204A** may correspond to a plurality of transaction records that may be associated with information, such as, ticketing information, subscription information, payment information, revenue sharing information, and/or mobility service information. Each of the plurality of transaction records may be recorded and stored in the first distributed ledger database **118**. The information related to the subscription **204B** may correspond to subscription information associated with users or riders. For example, the information related to the subscription **204B** may include information such as date of subscription, end date of subscription, type of subscription plan, and the like. The system for revenue sharing from history **204C** may be based on a plurality of transaction records stored on a first MP node or the first distributed ledger database **118**.

The system for GDPR compliant user auth **204D** may correspond to treatment of personally identifiable information (PII), as defined by government or compliance rules and policies. The GDPR may lay down certain guidelines that the organizations must follow which may limit how PII data can be used. The system for roaming auth **204E** may correspond to information associated with a travelling user (such as, a roaming user). The transaction records may further be associated with the roaming user. For example, the roaming user may be a rider whose trip may be such that a first leg of the trip may be associated with a first MaaS player and a first MP, while a second leg of the trip may be associated with a second MaaS player and a second MP. The plurality of transaction records for the roaming user may be associated with information, such as, ticketing information, subscription information, payment information, revenue sharing information, and/or mobility service information.

In an embodiment, the record history **204A**, the information related to the subscription **204B**, and the revenue sharing from history **204C** may correspond to a blockchain common database (DB). In an example, the blockchain common DB may be implemented as a set of nodes of the first distributed ledger database **118**. In an embodiment, the system for GDPR compliant user auth **204D** and the system for roaming auth **204E** may be used for authentication of nodes and users based on decentralized identifiers or dis-

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tributed identifiers (DIDs) and self-sovereign identities (SSI). The information related with the system for GDPR compliant user auth **204D** and the system for roaming auth **204E** may be stored as credential information in a decentralized identification (DID) blockchain database. In an example, the decentralized identification (DID) blockchain database may be implemented as a set of nodes of the second distributed ledger database **120**.

The system for real-time analysis and recommendation **204F** may provide a real-time analytics and recommendations based on one or more queries received from the set of nodes of the MaaS platform **108**. The system for real-time analysis and recommendation **204F** may store a first group of transaction records of a first plurality of transaction records associated with all the mobility providers of a first MaaS network (e.g., the MaaS network **100**). The storage of the first group of transaction records may be based on a first data retention threshold and a first storage duration of first plurality of transaction records on the first MaaS node. For example, if the first data retention threshold is three days, all transaction records of the first plurality of transaction records, that have a storage duration (i.e., first storage duration) of three or more days at the first MaaS node, may be selected as the first group of transaction records at the first MaaS node. In an embodiment, the system for real-time analysis and recommendation **204F** may be configured to store selected transaction records from the first group of transaction records, based on an aggregation logic. In an embodiment, the system for real-time analysis and recommendation **204F** may receive (from the electronic device **102** or a server associated with the MaaS platform **108**) one or more queries associated with the stored first group of transaction records.

The service options **206** may correspond to various services provided by the MaaS platform **108**. The service options **206** may include at least one of the route setting option **206A**, the system for tracking **206B**, the system for ticketing **206C**, the user information database **206D**, the payment information **206E**, and the information related to the user preference **206F**. The route setting option **206A** may correspond to a route that the MaaS platform **108** or the electronic device **102** selects for a trip of a rider. The trip may involve provision of a transportation service by multiple mobility providers, which may span multiple geographic regions. For example, the route may include a travel by train, bus, and taxi for travel from start point to destination point. The system for tracking **206B** may correspond to tracking of user ticket transaction data, travel data etc. associated with transaction records that may include, but is not limited to, transaction data corresponding to a trip-leg associated with a first MP, from ticket data in each of the first set of transaction records. The system for ticketing **206C** may correspond to a transit pass or a travel card that may be issued to users (such as passenger) that allows a passenger of the service to take either a predefined number of trips or unlimited trips for a predefined time-period. For example, the passenger may be issued a monthly transit pass or travel card to take unlimited number of trips for a time-period of one month. The user information database **206D** may be configured to store information associated with user (such as passenger). The information associated with the user may include, but is not limited to name, age, gender, travel history, and the like. The payment information **206E** may correspond to information associated with payment details and payment history of a user (such as passenger). The payment information may include, but is not limited to cost of ticket, amount spent within a predefined time-period,

mode of payment, amount due, and the like. The information related to the user preference **206F** may correspond to preferences associated with a user (such as a passenger). The information related to the user preference **206F** may correspond to preferences of a user (such as a passenger) associated with transportation. For example, the passenger may have a preference related to a mode of travel, for example, via a bus, as compared to other modes of travel, such as, a train, an airplane.

The additional service API **208** may correspond to an API that may help to integrate additional services associated with MaaS platform **108**. The additional service API **208** may include the LBS providers **208A**, the insurance service **208B**, and other services **208C**. the LBS providers **208A** may correspond to secondary services that may be based on location of the users. The LBS may correspond to services which utilize geographic information of the users to provide services or information to the users. For example, the LBS may include but is not limited to navigation service, advertisement service, location tracking service, location-based weather information service, and the like. The insurance service **208B** may correspond to an insurance service that may cover different travel related risks for a rider. The insurance service **208B** may include insurance cover for medical expenses, lost luggage, flight cancellations, and other losses that a traveler may incur in transit or during a trip. The other services **208C** may correspond to other travel related services that may include, but is not limited only to hotels, restaurants, and theme parks.

The API/onboarding agreement process **202B** may be used to integrate the second set of MaaS player nodes **106** with the MaaS layer **220**. The second set of MaaS player nodes **106** may be associated with the first set of MP nodes **104** with the help of the API/onboarding agreement process **202A** and API/onboarding agreement process **202B**. The first set of MP nodes **104** may be associated with a certain mobility provider and may be communicatively coupled to MP node(s) of the respective mobility providers. For example, a first MP node may be associated with a first MP (and coupled to the first MP node) and a second MP node may be associated with a second MP (and coupled to the second MP node). In some scenarios, a single MP node may be associated with each MP node (of that single MP), which may be included in multiple node packages. In certain embodiments, the on-boarding process of an MP node may be executed by a central MP node (e.g., the electronic device **102**) associated with the MP node, which may be associated with the same mobility provider as the MP node. In another embodiment, the on-boarding process of a MaaS player node may be executed by a central MaaS node (e.g., the electronic device **102**) associated with the MaaS player node, which may be associated with the same MaaS player as the MaaS player node.

The communication network **212** may include a communication medium through which the set of nodes of the MaaS network **100** may communicate with one another. The communication network **212** may be one of a wired connection or a wireless connection. Examples of the communication network **212** may include, but are not limited to, the Internet, a cloud network, Cellular or Wireless Mobile Network (such as, Long-Term Evolution and 5th Generation (5G) New Radio (NR)), a satellite communication system (using, for example, a network including a set of low earth orbit satellites), a Wireless Fidelity (Wi-Fi) network, a Personal Area Network (PAN), a Local Area Network (LAN), or a Metropolitan Area Network (MAN). Various devices in the network environment **100** may be configured

to connect to the communication network **212** in accordance with various wired and wireless communication protocols. Examples of such wired and wireless communication protocols may include, but are not limited to, at least one of a Transmission Control Protocol and Internet Protocol (TCP/IP), User Datagram Protocol (UDP), Hypertext Transfer Protocol (HTTP), File Transfer Protocol (FTP), Zig Bee, EDGE, IEEE 802.11, light fidelity (Li-Fi), 802.16, IEEE 802.11s, IEEE 802.11g, multi-hop communication, wireless access point (AP), device to device communication, cellular communication protocols, and Bluetooth (BT) communication protocols.

The layered architecture diagram **200** may further include the MaaS application interface (or MaaS API) **214**. The MaaS API **214** may be used to facilitate interaction of the set of users **216** with the MaaS network **100**. The set of users **216** may use a respective user device (not shown) to access nodes of the MaaS network **100**. Examples of the user devices may include, but are not limited to, a computing device, a desktop, a personal computer, a laptop, a computer workstation, a tablet computing device, a smartphone, a cellular phone, a mobile phone, a wearable display, a utility device, a utility console, a smart television (TV), an augmented reality (AR)/virtual reality (VR)/mixed reality (MR) device, a head-mounted display (HMD) for utility, a heads-up display (HUD) for utility, smart glasses for utility, a mainframe machine, a server, and/or a consumer electronic (CE) device.

FIG. 3 is an exemplary block diagram that illustrates an onboarding agreement system, in accordance with an embodiment of the disclosure. FIG. 3 is explained in conjunction with elements from FIGS. 1 and 2. With reference to FIG. 3, there is shown an onboarding agreement system **300** that may include the first set of MP nodes **104**, the second set of MaaS player nodes **106** of FIG. 1, and the set of users **216** of FIG. 2. The onboarding agreement system **300** may further include a MaaS community **302** and a Business-to-Business (B2B) white label platform provider **304**. The first set of MP nodes **104** and the second set of MaaS player nodes **106** may be communicatively coupled with the MaaS community **302**. Further, the MaaS community **302** may be communicatively coupled with the Business to Business (B2B) white label platform provider **304**. The set of users **216** may be offered multi-modal travel services of the MaaS platform **108**, through the second set of MaaS player nodes **106**.

The electronic device **102** may be configured to receive a set of onboarding agreements from a set of nodes of a MaaS network (such as the MaaS network **100**). The set of nodes may include at least one of a first set of mobility provider nodes (such as the first set of MP nodes **104**) and a second set of MaaS player nodes (such as the second set of MaaS player nodes **106**). The set of onboarding agreements may correspond to at least one of a master service agreement (such as the master service agreement), a terms-of-service agreement (such as the terms-of-service agreement), or a privacy policy agreement (such as the privacy policy agreement). In an example, the electronic device **102** may receive a service agreement and a privacy policy (PP) agreement from a MaaS player node (of the second set of MaaS player nodes **106**) and may receive a terms-of-service (ToS) agreement and a privacy policy agreement from an MP node (of the first set of MP nodes **104**). In another example, the electronic device **102** may receive a service agreement and a privacy policy agreement from an MP node (of the first set of MP nodes **104**) and may receive a terms-of-service agreement (ToS) and a privacy policy (PP) agreement from

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a MaaS player node (of the second set of MaaS player nodes **106**). A computing device of the set of users **216** may receive a service agreement, an end-user license agreement (EULA), and a privacy policy (PP) agreement from a MaaS player node associated with the MaaS player which offers a MaaS service to the respective user of the set of users **216**.

The electronic device **102** may generate a set of service and agreement templates (such as, the set of service and agreement templates) based on the received set of onboarding agreements to create a MaaS node group (such as, the MaaS community or coalition **302**) corresponding to the set of nodes. The set of service and agreement templates may be skeleton contract/agreement documents with pre-defined legal constructs for a certain use case or purpose (e.g., revenue sharing or data sharing). In an embodiment, the MaaS community **302** may be associated with the electronic device **102** (not shown in FIG. 3). For example, the MaaS community **302** may correspond to the MaaS node group created by the electronic device **102**. In an embodiment, the MaaS community **302** may correspond to a MaaS common harmonization system that may be operated by a public authority, a government organization, a municipality, or a MaaS business community (including for example, MPs and MaaS players).

The electronic device **102** may transmit the received set of onboarding agreements and the generated set of service and agreement templates to the created MaaS node group (such as, the MaaS community **302**). In an embodiment, the MaaS community **302** may generate the MSA, the TOS, and the PP. For example, the generated set of service and agreement templates may be skeletal formats that may be generated for certain service agreements etc. For example, if two organizations have to establish a revenue share service, the organization may be required to establish a service agreement between the organizations for revenue sharing. In an embodiment, the agreement template may be generated by the MaaS community **302**.

In an embodiment, the B2B white label platform provider **304** may be a B2B service provider that may offer an online portal to provide a templated travel solution to other businesses based on configurable MaaS services, through the MaaS network **100**. In certain scenarios, the MaaS community **302** may receive a service agreement and a privacy policy agreement from the B2B white label platform provider **304**. The received service agreement and privacy policy may be used by the MaaS community **302** to generate templated service agreements, such as, service agreements for revenue sharing and service agreements associated with various subscription models.

The electronic device **102** may receive an onboarding request from a first node (for example, a MaaS player node of the second set of MaaS player nodes **106**) of the MaaS network **100**. In an example, the onboarding request may be received from the set of users **216** through the second set of MaaS player nodes **106**. The electronic device **102** may further verify a DID of the first node based on the received onboarding request. For example, by using DID and Self Sovereign Identity (SSI), the electronic device **102** may verify the credentials of the first node. It may be appreciated that by verification of the credentials of the first node, an identity of the user or personnel associated with an organization may be authenticated. Further, based on verification of the first node, information associated with the user or the personnel associated with the organization may be stored at an identity provider (IDP). In addition, based on verification

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of the first node, an access to the MaaS platform **108** may be granted to the user or the personnel associated with the organization.

The electronic device **102** may be configured to transmit, to the first node, a set of parameters and a service agreement associated with the first node, based on the verified DID, transmitted set of onboarding agreements, and the set of service and agreement templates. For example, based on the granted access, the set of parameters, and the service agreement, the user or the personnel may generate service agreement by using the set of service and agreement templates that may have been generated by the MaaS community **302**.

The electronic device **102** may transmit the service agreement associated with the first node to the created MaaS node group (such as, the MaaS community **302**). Thereafter, the electronic device **102** may generate an API key of a data hub for the MaaS network **100**, based on the received service agreement. The first node may be onboarded to the created MaaS node group (such as, the MaaS community **302**) of the MaaS network **100** based on the generated API key and the transmitted service agreement associated with the first node. For example, the service agreements and the shared API keys may enable the first node to access the data hub of the MaaS network **100** and communicate with the other nodes of the MaaS network **100**, and offer or consume MaaS services.

FIGS. 4A and 4B illustrate exemplary architecture for credential request and verification of a node to be onboarded to a MaaS network, in accordance with an embodiment of the disclosure. FIGS. 4A and 4B are explained in conjunction with elements from FIG. 1, FIG. 2, and FIG. 3. With reference to FIGS. 4A and 4B, there is shown an exemplary architecture **400** for credential request and verification of a node to be onboarded to a MaaS network (e.g., the MaaS network **100**). The architecture **400** may include the middleware **110**, the issuer node **112**, the set of holder nodes **114**, the verifier node **116**, a web application **408**, a webhook **410**, and a Cognito user pool **418**. In an embodiment, the middleware **110** may include the webhook **410** and an API **412**. In another embodiment, the middleware **110** may include the webhook **410**, an onboarding request database **414**, a user database **414A**, and a scheduler **416**. The verifier node **116** may include a DID agent **402A**. The set of holder nodes **114** may include a DID agent **402B**. The issuer node **112** may include a DID agent **402C**, a set of attributes **404**, and a set of credentials **406**. The web application **408** may include a web UI **408A**.

Each of the issuer node **112**, the set of holder nodes **114**, and the verifier node **116** may be associated with the second distributed ledger database **120** and may include a corresponding DID agent. The DID agent **402C** of the issuer node **112** may generate one or more credentials or credential templates that may be stored as the set of credentials **406**. Further, the DID agent **402C** of the issuer node **112** may define attributes that may correspond to the generated credentials. The attributes may be stored as the set of attributes **404**. The set of attributes **404** may include, but is not limited to, an organization ID **404A**, an organization name **404B**, a name **404C** (of an administrator user), an electronic-mail ID **404D** (of an administrator user), a role **404E** (of a user of a node), and a user privilege level **404F** (of a user of a node). The issuer node **112** may be configured to generate credentials associated with each holder node of the set of holder nodes **114**. The issuer node **112** may correspond to a government organization or a private organization such as, an entity associated with the MaaS community **302**, that may issue credentials for nodes of the MaaS network **100**.

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With reference to FIG. 4A, the web UI **408A** of the web application **408** may be configured to receive a user input from a user or a personnel of an organization to raise a request for onboarding **408B** (of a node, such as, a first node). The user input may include any of selection of an option on the web UI **408A**, or scan of a quick response (QR) code. The web application **408** may be accessed by the user or the personnel using a user device (such as, mobile device). The API **412** may receive the user input (such as, a QR code **412A**) and information associated with the user device. The information associated with the user device may include type of device, type of connection, and the like. Further, the API **412** may process QR code **412A** and submit a connection request **412C** (including information, such as, connection details **412B**) to the verifier node **116** through the middleware **110**. The verifier node **116** may share an invitation for connection with the DID agent **402B** of a relevant holder node (of the set of holder nodes **114**) and receive a connection identifier, a DID, and a label associated with the first node.

The verifier node **116** may be configured to receive a credential verification request (e.g., a credential verification request **410A**) associated with the first node, based on the received request for onboarding **408B** of the first node. For example, based on the received connection identifier, the verifier node **116** may trigger (for example, an event associated with a credential verification request **410A**) the webhook **410** to automatically transmit a request for credential verification to the verifier node **116**. The webhook **410** may correspond to a call-back function or sub-routine for event triggered communication between applications. For example, the webhook **410** may correspond to a service that enables one node to send data to another node based on a triggering event. In current scenarios, the verifier node **116** may trigger an event (associated with the credential verification request **410A**) on the webhook **410** based on the reception of the connection identifier by the verifier node **116**. Based on the event triggered on the webhook **410**, the webhook **410** may automatically transmit a credential verification request **410B** (refer FIG. 4B) to the verifier node **116**.

With reference to FIG. 4B, the verifier node **116** may be configured to query a holder node of the set of holder nodes **114** for verified credentials of the first node, based on the credential verification request **410B** received from the webhook **410**. Further, the verifier node **116** may be configured to receive, from the holder node, the verified credentials of the first node, based on the query. The DID of the first node may be verified based on the received verified credentials of the first node. Further, based on the verification of the first node, the first node may be onboarded to the created MaaS node group of the MaaS network **100**. Further, based on the verified credentials of the first node, the verifier node **116** may be configured to trigger an event of onboarding of the first node, via the webhook **410**. In an embodiment, the onboarding may further include generation of access or credentials for an identity provider (IDP) (such as, a Cognito user from a Cognito user pool **418**). The IDP may correspond to a service that may be configured to store and manage identity, credentials etc., of users or personnel of an organization. The organizations may use the IDP service to allow personnel associated with the organization to connect to resources that may be necessary. The Cognito user of the Cognito user pool **418** may be configured to create a user profile and assign role (as denoted by **416A**) of a new user. The onboarding request database **414** and the user database **414A** may be used for storage of information associated with

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the created user profile and respective assigned role, and information associated with the onboarding of the first node.

The scheduler **416** may include suitable logic, circuitry, interfaces, and/or code that may be configured to generate a trigger event associated with creation of the user profile and assignment of the role (as denoted by **416A**). In an embodiment, the trigger event may include verification of credentials of the first node. Examples of the scheduler **416** may include, but are not limited to, a smartphone, a cellular phone, a mobile phone, a mainframe machine, a server, a computer workstation, and/or any computing device with a schedule-based or an event-based triggering mechanism. In an embodiment, the electronic device **102** may further be configured to create a user profile and a role for the user profile, associated with the first node in the MaaS network **100**. The creation of the user profile and the role may be based on the verified DID of the first node and the scheduler **416**.

FIG. 5 is an exemplary sequence diagram that illustrates establishment of a master service agreement (MSA) between nodes of a Mobility-as-a-Service (MaaS) platform, in accordance with an embodiment of the disclosure. FIG. 5 is explained in conjunction with elements from FIGS. 1, 2, 3, 4A, and 4B. With reference to FIG. 5, there is shown a sequence diagram **500** that illustrates a sequence of operations from **506** to **512**. The sequence of operations may be executed by various nodes of the MaaS network **100** (such as, a first MaaS player node **502** of the second set of MaaS player nodes **106**, the first mobility provider node **504** of the first set of MP nodes **104** of FIG. 5, and the electronic device **102** of FIG. 1).

At **506**, a first master service agreement (MSA) and counterparty details may be received. The electronic device **102** may be configured to receive, from the first MaaS player node **502** of the second set of MaaS player nodes **106**, the first master service agreement (MSA) and counterparty details associated with the first MaaS player node **502**. The counterparty details may correspond to the first mobility provider node **504** of the first set of mobility provider nodes **104**. In an embodiment, for onboarding a node on the MaaS network **100**, the MSA may be established by the MaaS community **302**. For example, the MSA may correspond to agreements that may need to be established between organizations that needs to share services such as, revenue, data, and logistic services.

At **508**, the first MSA may be transmitted. The electronic device **102** may be configured to transmit, to the first mobility provider node **504**, the first MSA, based on received the counterparty details. In an embodiment, the first MSA may be received and accepted by both the first MaaS player node **502** and the first mobility provider node **504** to initiate the services such as, data sharing, revenue sharing, and logistic services. In an embodiment, the MSA may be received from the first MaaS player node **502**. The counterparty details may indicate that the MSA may be between a first MaaS player (associated with the first MaaS player node **502**) and a first MP (associated with the first mobility provider node **504**). The first MaaS player may be a first party and the first MP may be a counterparty to the MSA.

At **510**, an input indicative of an acceptance of the transmitted first MSA may be received. The electronic device **102** may be configured to receive, from the first mobility provider node **504**, the input indicative of an acceptance of the transmitted first MSA. For example, a user of the first mobility provider node **504** may provide an input indicative of acceptance or rejection of the first MSA.

At **512**, the electronic device **102** may be configured to establish the first MSA between the first MaaS player node **502** and the first mobility provider node **504**, based on the received input indicative of the acceptance of the transmitted first MSA. For example, if the first MaaS player and the first mobility provider intends to initiate a revenue share service, then both the first MaaS player and the first mobility provider may be required to receive and accept the first MSA. For example, the first MSA may facilitate an establishment of a legally-bound service contract between the first MaaS player and the first mobility provider based on an acceptance of the first MSA by both parties (i.e., the first MaaS player and the first mobility provider) of the MSA. Based on the establishment of the first MSA between both the parties (i.e., the first MaaS player and the first mobility provider), the parties may establish service agreements, and terms-of-service policies, to get on-boarded to the MaaS platform **108**, as described further, for example, in FIGS. **6**, **7**, and **8**.

FIG. **6** is an exemplary sequence diagram that illustrates establishment of a service agreement between nodes of a Mobility-as-a-Service (MaaS) platform, in accordance with an embodiment of the disclosure. FIG. **6** is explained in conjunction with elements from FIGS. **1**, **2**, **3**, **4A**, **4B**, and **5**. With reference to FIG. **6**, there is shown a sequence diagram **600** that illustrates a sequence of operations from **602** to **610**. The sequence of operations may be executed by various nodes of the MaaS network **100** (such as, the first MaaS player node **502**, the first mobility provider node **504** of FIG. **5**, and the electronic device **102** of FIG. **1**).

At **602**, a first service agreement may be received. The electronic device **102** may be configured to receive, from the first MaaS player node **502**, the first service agreement associated with the first MaaS player node **502**. In an embodiment, a service agreement may correspond to a contract that may define terms and conditions of a services or revenue share arrangement between two or more organizations (such as, a MaaS player and a mobility provider).

At **604**, the first service agreement and a process service may be associated. The electronic device **102** may be configured to associate the received first service agreement with process service information and the first MSA, to determine an MSA bound service agreement. For example, the first service agreement may correspond to contract or agreement between two parties, such as, the first MaaS player node **502** and the first mobility provider node **504**, for MaaS services to be offered and a revenue sharing arrangement between the two parties. Further, the process service may correspond to processes or services that may be offered by the first MaaS player node **502** and the first mobility provider node **504**. The first service agreement and process service may be associated with the MSA to determine the MSA bound service agreement. For example, the MSA-bound service agreement may correspond to the acceptance of the terms of agreement of the first MSA, as generated by the MaaS community **302**, and service agreements as generated between the first MaaS player node **502** and the first mobility provider node **504**.

At **606**, an MSA-bound service agreement may be transmitted. The electronic device **102** may be configured to transmit, to the first mobility provider node **504**, the determined MSA-bound service agreement. The MSA-bound service agreement may be received by the first mobility provider node **504**.

At **608**, an input indicative of an acceptance of the transmitted MSA-bound service agreement may be received. The electronic device **102** may be configured to receive,

from the first mobility provider node **504**, an input indicative of an acceptance of the transmitted MSA-bound service agreement. For example, the first mobility provider node **504** may accept the MSA-bound service agreement based on a user input indicative of the acceptance. Based on the acceptance of the MSA-bound service agreement, terms of the MSA-bound agreement may be enforceable.

At **610**, the first service agreement may be established. The electronic device **102** may be configured to establish the first service agreement between the first MaaS player node **502** and the first mobility provider node **504**, based on the received input indicative of the acceptance of the transmitted MSA-bound service agreement. For example, once parties such as, the first MaaS player node **502** and the first mobility provider node **504**, have accepted the MSA-bound service agreement, the service agreement may be established between the parties and services such as, revenue sharing and data sharing, may be provisioned between the parties.

FIG. **7** is an exemplary sequence diagram that illustrates establishment of a first terms of service (ToS) between nodes of a Mobility-as-a-Service (MaaS) platform, in accordance with an embodiment of the disclosure. FIG. **7** is explained in conjunction with elements from FIG. **1**, FIG. **2**, FIG. **3**, FIG. **4A**, FIG. **4B**, FIG. **5**, and FIG. **6**. With reference to FIG. **7**, there is shown a sequence diagram **700** that illustrates a sequence of operations from **702** to **710**. The sequence of operations may be executed by various nodes of the MaaS network **100** (such as, the first MaaS player node **502**, the first mobility provider node **504** of FIG. **5**, and the electronic device **102** of FIG. **1**).

At **702**, a first terms of service (ToS) may be received. The electronic device **102** may be configured to receive, from the first MaaS player node **502**, a first terms-of-service (ToS) associated with the first MaaS player node **502**. The ToS may correspond to legal agreements between the first mobility provider node **504** and the first MaaS player node **502**. Both the first MaaS player node **502** and the first mobility provider node **504** may be required to agree to abide by the terms of service for the enforcement of a related service agreement (e.g., the first service agreement).

At **704**, the received first ToS and the first service agreement may be associated. The electronic device **102** may be configured to associate the received first ToS with the first service agreement to determine a service agreement bound-ToS. The first MaaS player node **502** and the first mobility provider node **504** may agree to abide by the ToS and the first service agreement, to establish the ToS between a first MaaS player (associated with the first MaaS player node **502**) and a first MP node (associated with the first MP node **504**).

At **706**, the determined service agreement bound-ToS may be transmitted. The electronic device **102** may be configured to transmit, to the first mobility provider node **504**, the determined service agreement bound-ToS. For example, the first mobility provider node **504** may accept the service agreement bound-ToS based on an input indicative of an acceptance of the ToS. Based on acceptance of the service agreement bound-ToS, the terms of the service agreement bound-ToS may be enforceable.

At **708**, an input indicative of an acceptance of the transmitted service agreement bound-ToS may be received. The electronic device **102** may be configured to receive, from the first mobility provider node **504**, an input indicative of an acceptance of the transmitted service agreement bound-ToS.

At **710**, the first ToS may be established between the first MaaS player node **502** and the first mobility provider node

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504. The electronic device 102 may be configured to establish the first ToS between the first MaaS player node 502 and the first mobility provider node 504, based on the received input indicative of the acceptance of the transmitted service agreement bound-ToS. Thus, the first ToS may be established between a first MaaS player (associated with the first MaaS player node 502) and a first MP (associated with the first mobility provider node 504). Based on the establishment of the first ToS, the first MaaS player node 502 and the first mobility provider node 504 may communicate with each other and other nodes of the MaaS platform 108, to offer or consume MaaS services associated with the MaaS platform 108.

FIG. 8 is an exemplary sequence diagram that illustrates establishment of a second terms of service (ToS) between nodes of a Mobility-as-a-Service (MaaS) platform to override a previous version of ToS established between the nodes, in accordance with an embodiment of the disclosure. FIG. 8 is explained in conjunction with elements from FIGS. 1, 2, 3, 4A, 4B, 5, 6, and 7. With reference to FIG. 8, there is shown a sequence diagram 800 that illustrates a sequence of operations from 802 to 808. The sequence of operations may be executed by various nodes of the MaaS network 100 (such as, the first MaaS player node 502, the first mobility provider node 504 of FIG. 5, and the electronic device 102 of FIG. 1).

At 802, a second ToS may be received. The electronic device 102 may be configured to receive, from the first MaaS player node 502, a second ToS associated with the first MaaS player node 502. The second ToS may correspond to a new version of the established first ToS.

At 804, the second ToS may be transmitted. The electronic device 102 may be configured to transmit, to the first mobility provider node 504, the received second ToS.

At 806, an input indicative of an acceptance of the second ToS may be received. The electronic device 102 may be configured to receive, from the first mobility provider node 504, an input indicative of an acceptance of the second ToS. For example, the input indicative of the acceptance of the second ToS may be received as a user input, via the first mobility provider node 504.

At 808, the first ToS established between the first MaaS player node 502 and the first mobility provider node 504 may be disabled. The electronic device 102 may be configured to disable the first ToS established between the first MaaS player node 502 and the first mobility provider node 504, based on the received input indicative of the acceptance of the second ToS.

At 810, the second ToS may be established. The electronic device 102 may be configured to establish the second ToS between the first MaaS player node 502 and the first mobility provider node 504, based on the disablement of the first ToS. Thus, once the first ToS (i.e., a previous version of ToS) established between two parties, such as, the first MaaS player node 502 and the first mobility provider node 504 is disabled, the electronic device 102 may establish the second ToS (i.e., a new version of ToS) between the parties, based on the acceptance of the second ToS by both the parties. The newly established second ToS may override a previous version of ToS (e.g., the first ToS). For example, the terms of service between various parties (e.g., MaaS players and mobility providers) of the MaaS platform 108 may be updated easily from time-to-time, based on various requirements, such as, legal, contractual, or service requirements.

FIG. 9 is an exemplary sequence diagram that illustrates secure onboarding of users and organizations on a mobility-as-a-service (MaaS) platform, in accordance with an

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embodiment of the disclosure. FIG. 9 is explained in conjunction with elements from FIGS. 1, 2, 3, 4A, 4B, 5, 6, 7 and 8. With reference to FIG. 9, there is shown a sequence diagram 900 that illustrates a sequence of operations from 910 to 936. The sequence of operations may be executed by various nodes of the MaaS network 100 (such as, a holder-agent node 902, a verifier-agent node 904, a verifier-middleware node 906, a verifier-web UI node 908 of FIG. 9, and the set of holder nodes 114 of FIG. 1).

At 910, invitation details may be generated. In an embodiment, the set of holder nodes 114 may be configured to generate the invitation details. The invitation details may correspond to onboarding of a first node to the MaaS network 100. The generated invitation details may be transmitted to the holder-agent node 902.

At 912, the generated invitation details may be submitted. In an embodiment, the set of holder nodes 114 may submit the invitation details. The submitted invitation details may be transmitted to the verifier-webUI node 908, for example, a platform owner (such as a MaaS operator).

At 914, the invitation details may be propagated. In an embodiment, the verifier-webUI node 908 may propagate the invitation details through the verifier-middleware node 906. The verifier-webUI node 908 may transmit the invitation details to the verifier-agent node 904 through the verifier-middleware node 906.

At 916, based on the reception of the propagated invitation details, a connection may be established between the holder-agent node 902 and the verifier-agent node 904. In an example, the connection may be a synchronous serial interface (SSI) connection.

At 918, based on the connection established between the holder-agent node 902 and the verifier-agent node 904, a connection to the webhook 410 may be triggered. In an embodiment, the webhook 410 may be triggered by the verifier-agent node 904. To trigger the connection, the verifier-agent node 904 may transmit a webhook connection update to the verifier-middleware node 906.

At 920, a proof request may be initiated. The proof request may be transmitted from the verifier-middleware node 906 to the verifier-agent node 904. In an embodiment, the proof request may be initiated for verification of credentials associated with the verifier-webUI node 908.

At 922, credentials may be selected for verification and the selected credentials may be attached to the proof request. In an embodiment, the set of holder nodes 114 may select the credentials for verification. Further, the selected credentials may be attached to the proof request. The proof request may be transmitted from the set of holder nodes 114 to the holder-agent node 902.

At 924, the updated proof request may be transmitted from the holder-agent node 902 to the verifier-agent node 904. The holder-agent node 902 may transmit (or forward) the updated proof request (i.e., the proof request including the selected credentials) to the verifier-agent node 904.

At 926, based on the reception of the updated proof request at the verifier-agent node 904, the webhook 410 may be triggered. The verifier-agent node 904 may trigger the webhook 410 based on the reception of the updated proof request. Based on trigger of the webhook 410, a webhook proof request update may be transmitted from the verifier-agent node 904 to the verifier-middleware node 906.

At 928, a proof verification request may be initiated. The verifier-middleware node 906 may initiate the proof verification request. The credentials of the verifier-webUI node 908 may be verified based on the received proof request.

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At **930**, a proof request may be verified against its validity and custom restrictions. The verifier-agent node **904** may verify the proof request (i.e., the proof verification request) based on validity of the proof request and custom restrictions, such as, a decentralized/distributed ID (DID) of an issuer node.

At **932**, a response may be transmitted. The verifier-agent node **904** may transmit the response to the holder-agent node **902**. The response may correspond to the updated proof request transmitted (at **924**) by the holder-agent node **902**.

At **934**, a webhook proof update request update may be transmitted. The webhook proof update request may be transmitted from the verifier-agent node **904** to the verifier-middleware node **906**, based on the verification of the credentials.

At **936**, based on verification and the webhook proof update, if the proof is determined as valid, the first node may be onboarded. The verifier-middleware node **906** may determine whether the proof (associated with the webhook proof update) is valid, based on the verification of the credentials (at **934**). In case the proof is determined as valid, the verifier-middleware node **906** may retrieve (or read) whitelisted attributes from the attached credentials (included, for example, in the webhook proof update). The whitelisted attributes and the attached credentials may be stored in a local database associated with a data owner (for example, an owner of the first node). Thus, private information or sensitive details of the data owners may be stored locally thereby maintaining privacy and confidentiality of the data owners. Based on the validity of the credentials of the first node, the first node may be onboarded on the MaaS platform **108**. The first node may receive an API key related to a data hub associated with the MaaS community **302**, based on the onboarding of the first node on the MaaS platform **108**. Further, based on the onboarding, the first node may be communicably coupled to other nodes of the MaaS network **100** and may participate in MaaS transactions to offer and/or consume MaaS services, through the MaaS platform **108**. Typically, for a MaaS platform, there may be no streamlined process to onboard the various entities, such as, users, organizations, and personnel associated with organizations. The onboarding of a new user or a new organization to a MaaS platform may be a cumbersome and a time-consuming process that may require manual verification or authorization. For example, manual verification of new users through email may be required. In such case, the onboarding process may be an ad-hoc process. During the onboarding process, sensitive information (such as, name, email address, or organization name etc.) associated with the user, the organization, or the personnel associated with the organization may be available to third party administrators of the MaaS platform. The availability of such sensitive information to the third parties may give rise to confidentiality and privacy concerns.

The disclosed electronic device **102** may receive an onboarding request from a first node (such as the first node) of the MaaS network **100** (or the MaaS platform **108**). Based on the received onboarding request, a decentralized identity (DID) (such as the DID) of the first node may be verified. Further, based on the verified DID, transmitted set of onboarding agreements, and the set of service and agreement templates, a set of parameters and a service agreement associated with the first node may be transmitted to the first node. Further, the electronic device **102** may transmit the service agreement associated with the first node to the created MaaS node group (e.g., the MaaS community **302**). Thereafter, based on the received service agreement, an

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Application Programming Interface (API) key of a data hub for the MaaS network **100** may be generated. Thus, the first node may be onboarded to the created MaaS node group of the MaaS network **100** based on the generated API key and the transmitted service agreement associated with the first node. For example, the first node, that may correspond to any computing device of a user, an organization, or personnel associated with the organization, may be onboarded based on generation of the API key and transmission of the service agreements. Further, the API key may be used by the onboarded user (or the organization, or personnel associated with the organization) to share data (such as, revenue data, travel data, cyclic revenue, or travel data). Thus, a streamlined and secure on-boarding of new nodes to the MaaS platform may be achieved without concerns related to confidentiality and privacy of various entities (such as, the users/organizations/personnel) to be on-boarded, as the private information or sensitive information of data owners (i.e., users of the entities) may only be stored locally. After an entity is successfully on-boarded to the MaaS platform, the entity may start to communicate with the other entities of the MaaS platform **108** and offer service (for example, in case of MaaS player nodes or mobility provider nodes) or consume services (for example, in case of end-users or mobility provider nodes) associated with the MaaS platform **108**.

FIG. **10** is a block diagram that illustrates an exemplary electronic device of FIG. **1**, in accordance with an embodiment of the disclosure. FIG. **10** is explained in conjunction with elements from FIG. **1**, FIG. **2**, FIG. **3**, FIG. **4A**, FIG. **4B**, FIG. **5**, FIG. **6**, FIG. **7**, FIG. **8**, and FIG. **9**. With reference to FIG. **10**, there is shown the exemplary electronic device **102**. The electronic device **102** may include circuitry **1002**, a memory **1004**, an input/output (I/O) device **1006**, and a network interface **1008**. The input/output (I/O) device **1006** may include a display device **1006A**. The network interface **1008** may connect the electronic device **102** with the first distributed ledger database **118** and the MaaS platform **108**, via the communication network **212**.

The circuitry **1002** may include suitable logic, circuitry, and/or interfaces that may be configured to execute program instructions associated with different operations to be executed by the electronic device **102**. The operations may include reception of the set of onboarding agreements, generation of the set of service and agreement templates, transmission of the received set of onboarding agreements and the generated set of service and agreement templates, reception of the onboarding request, verification of the DID, transmission of the set of parameters and the service agreement, transmission of the service agreement and generation of the API key. The circuitry **1002** may include one or more processing units, which may be implemented as a separate processor. In an embodiment, the one or more processing units may be implemented as an integrated processor or a cluster of processors that perform the functions of the one or more specialized processing units, collectively. The circuitry **1002** may be implemented based on a number of processor technologies known in the art. Examples of implementations of the circuitry **1002** may be an X86-based processor, a Graphics Processing Unit (GPU), a Reduced Instruction Set Computing (RISC) processor, an Application-Specific Integrated Circuit (ASIC) processor, a Complex Instruction Set Computing (CISC) processor, a microcontroller, a central processing unit (CPU), and/or other control circuits.

The memory **1004** may include suitable logic, circuitry, interfaces, and/or code that may be configured to store one or more instructions to be executed by the circuitry **1002**.

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The one or more instructions stored in the memory **1004** may execute different operations of the circuitry **1002** (and/or the electronic device **102**). The memory **1004** may be configured to store information related to the set of nodes. Further, the memory **1004** may store credential information and information related to the second set of MaaS player nodes **106**, the MaaS platform **108**, the issuer node **112**, the set of holder nodes **114**, the verifier node **116**, and each of the first distributed ledger database **118**. Examples of implementation of the memory **1004** may include, but are not limited to, Random Access Memory (RAM), Read Only Memory (ROM), Electrically Erasable Programmable Read-Only Memory (EEPROM), Hard Disk Drive (HDD), a Solid-State Drive (SSD), a CPU cache, and/or a Secure Digital (SD) card.

The I/O device **1006** may include suitable logic, circuitry, interfaces, and/or code that may be configured to receive an input and provide an output based on the received input. For example, the I/O device **1006** may receive an input indicative of acceptance of the first MSA, the MSA-bound service agreement, the service agreement bound-ToS, and the second ToS. The I/O device **1006** may be further configured to display a user-interface associated with the onboarding of a node (e.g., the first node) on the MaaS platform **108**. The I/O device **1006** may include various input and output devices, which may be configured to communicate with the circuitry **1002**. The I/O device **1006** may include the display device **1006A**. Examples of the I/O device **1006** may include, but are not limited to, a touch screen, the display device **1006A**, a keyboard, a mouse, a joystick, a microphone, or a speaker. Examples of the I/O device **1006** may further include braille I/O devices, such as, braille keyboards and braille readers.

The network interface **1008** may include suitable logic, circuitry, interfaces, and/or code that may be configured to facilitate communication between the electronic device **102**, the first distributed ledger database **118**, and the MaaS platform **108**. The network interface **1008** may be implemented by use of various known technologies to support wired or wireless communication of the electronic device **102** with the communication network. The network interface **1008** may include, but is not limited to, an antenna, a radio frequency (RF) transceiver, one or more amplifiers, a tuner, one or more oscillators, a digital signal processor, a coder-decoder (CODEC) chipset, a subscriber identity module (SIM) card, or a local buffer circuitry.

The network interface **1008** may be configured to communicate via wireless communication with networks, such as the Internet, an Intranet, a wireless network, a cellular telephone network, a wireless local area network (LAN), or a metropolitan area network (MAN). The wireless communication may be configured to use one or more of a plurality of communication standards, protocols and technologies, such as Global System for Mobile Communications (GSM), Enhanced Data GSM Environment (EDGE), wideband code division multiple access (W-CDMA), Long Term Evolution (LTE), 5th Generation (5G) New Radio (NR), code division multiple access (CDMA), time division multiple access (TDMA), Bluetooth, Wireless Fidelity (Wi-Fi) (such as IEEE 802.11a, IEEE 802.11b, IEEE 802.11g or IEEE 802.11n), voice over Internet Protocol (VoIP), light fidelity (Li-Fi), Worldwide Interoperability for Microwave Access (Wi-MAX), a protocol for email, instant messaging, and a Short Message Service (SMS).

The display device **1006A** may include suitable logic, circuitry, and interfaces that may be configured to display the web UI **408A**. The display device **1006A** may be a touch screen which may enable a user to provide a user-input via

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the display device **1006A**. The touch screen may be at least one of a resistive touch screen, a capacitive touch screen, or a thermal touch screen. The display device **1006A** may be realized through several known technologies such as, but not limited to, at least one of a Liquid Crystal Display (LCD) display, a Light Emitting Diode (LED) display, a plasma display, or an Organic LED (OLED) display technology, or other display devices. In accordance with an embodiment, the display device **1006A** may refer to a display screen of a head mounted device (HMD), a smart-glass device, a see-through display, a projection-based display, an electro-chromic display, or a transparent display. Various operations of the circuitry **202** for secure onboarding of users and organizations on a mobility-as-a-service (MaaS) platform are described further, for example, in FIGS. **3**, **4A**, **4B**, **5**, **6**, **7**, **8**, and **9**.

FIG. **11** illustrates an exemplary flowchart of a method for secure onboarding of users and organizations on a mobility-as-a-service (MaaS) platform, in accordance with an embodiment of the disclosure. FIG. **11** is described in conjunction with elements from FIGS. **1**, **2**, **3**, **4A**, **4B**, **5**, **6**, **7**, **8**, **9** and **10**. With reference to FIG. **11**, there is shown a flowchart **1100**. The exemplary method of the flowchart **1100** may be executed by any computing system, for example, by the electronic device **102** of FIG. **1**, the first set of MP nodes **104** or other MP nodes (e.g., a central MP node) of FIG. **1**, and/or the second set of MaaS player nodes **106** or other MaaS player nodes (e.g., a central MaaS player node) of FIG. **1**. The exemplary method of the flowchart **1100** may start at **1102** and proceed to **1104**.

At **1104**, the set of onboarding agreements may be received from a set of nodes of a mobility-as-a-service (MaaS) network. The circuitry **1002** may be configured to receive a set of onboarding agreements from a set of nodes of a mobility-as-a-service (MaaS) network (e.g., the MaaS network **100**). The set of nodes may include at least one of a first set of mobility provider nodes (such as the first set of MP nodes **104**) and a second set of MaaS player nodes (such as the second set of MaaS player nodes **106**). The set of onboarding agreements may correspond to at least one of a master service agreement (such as the master service agreement), a terms-of-service agreement (such as the terms-of-service agreement), or a privacy policy agreement (such as the privacy policy agreement). In an example, the set of onboarding agreements may be received from the second set of MaaS player nodes **106**. The MaaS network **100** may further include at least one of an issuer node (such as, the issuer node **112**) associated with a credential issuing authority, a set of holder nodes (such as, the set of holder nodes **114**) corresponding to the set of nodes, a verifier node (such as, the verifier node **116**) associated with a MaaS network (such as, the MaaS network **100**) owner, and distributed ledger databases (such as, the first distributed ledger database **118** and the second distributed ledger database **120**).

At **1106**, the set of service and agreement templates may be generated based on the received set of onboarding agreements to create a MaaS node group corresponding to the set of nodes. The circuitry **1002** may be configured to generate a set of service and agreement templates (such as, the set of service and agreement templates) based on the received set of onboarding agreements to create a MaaS node group (such as, the MaaS community **302**) corresponding to the set of nodes. The generation of the set of service and agreement templates and creation of the MaaS node group are described further, for example, in FIGS. **1** and **3**.

At **1108**, the received set of onboarding agreements and the generated set of service and agreement templates may be

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transmitted to the created MaaS node group. The circuitry **1002** may be configured to transmit the received set of onboarding agreements and the generated set of service and agreement templates to the created MaaS node group (such as, the MaaS community **302**). In an embodiment, the MaaS community **302** may generate the MSA, the ToS, and the PP. For example, the generated set of service and agreement templates may be skeletal formats that may be generated for certain service agreements etc. For example, if two organizations are to establish a revenue share service, the organizations may be required to establish a service agreement between the organizations for revenue sharing. In an embodiment, the agreement templates may be generated by the MaaS community **302**.

At **1110**, an onboarding request may be received from the first node of the MaaS network **100**. The circuitry **1002** may be configured to receive an onboarding request from a first node (for example, the second set of MaaS player nodes **106**) of the MaaS network **100**. In an example, the onboarding request may be received from the set of users **216** through the second set of MaaS player nodes **106**.

At **1112**, the DID of the first node may be verified based on the received onboarding request. The circuitry **1002** may be configured to verify a DID of the first node based on the received onboarding request. For example, by using DID and Self Sovereign Identity (SSI) of the first node, the electronic device **102** may verify the credentials of the first node. It may be appreciated that by verification of the credentials of the first node, an identity of a user or personnel associated with an organization associated with the first node may be authenticated. Further, based on verification of the first node, information associated with the user or the personnel associated with the organization may be stored at an identity provider (IDP). For example, based on verification of the first node, an access to the MaaS platform **108** may be granted to the user or the personnel associated with the organization. The verification of the credentials of the first node is described further, for example, in FIGS. **4A** and **4B**.

At **1114**, the set of parameters and the service agreement associated with the first node may be transmitted to the first node based on the verified DID, transmitted set of onboarding agreements, and the set of service and agreement templates. The circuitry **1002** may be configured to transmit, to the first node, a set of parameters and a service agreement associated with the first node, based on the verified DID, transmitted set of onboarding agreements, and the set of service and agreement templates. For example, based on the granted access, the set of parameters, and the service agreement, the user or the personnel of the organization may generate a service agreement. The generation of the service agreement may be further based on the set of service and agreement templates generated by the MaaS community **302**.

At **1116**, the service agreement associated with the first node may be transmitted to the created MaaS node group. The circuitry **1002** may be configured to transmit the service agreement associated with the first node to the created MaaS node group (such as, the MaaS community **302**).

At **1118**, the API key of the data hub for the MaaS network **100** may be generated based on the transmitted service agreement, wherein the first node may be onboarded to the created MaaS node group of the MaaS network **100** based on the generated API key and the transmitted service agreement associated with the first node. The circuitry **1002** may be configured to generate an API key of a data hub for the MaaS network **100**, based on the received service agreement. The

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first node may be onboarded to the created MaaS node group (such as, the MaaS community **302**) of the MaaS network **100** based on the generated API key and the transmitted service agreement associated with the first node. For example, based on the transmission of the service agreements if data may be needed to be shared, then the API keys may be generated using the DID credential. Further, the API keys may be shared with the first node to enable the first node to communicate with other nodes of the MaaS network **100** and for data sharing. The secure onboarding of the first node to the MaaS network **100** is described further, for example, in FIG. **9**. Control may pass to end.

Although the flowchart **1100** is illustrated as discrete operations, such as **1104**, **1106**, **1108**, **1110**, **1112**, **1114**, **1116** and **1118**, the disclosure is not so limited. Accordingly, in certain embodiments, such discrete operations may be further divided into additional operations, combined into fewer operations, or eliminated, depending on the particular implementation without detracting from the essence of the disclosed embodiments.

Various embodiments of the disclosure may provide a non-transitory computer-readable medium and/or storage medium having stored thereon, computer-executable instructions executable by a machine and/or a computer to operate an electronic device (for example, the electronic device **102**). Such instructions may cause the electronic device **102** to perform operations that may include reception of the set of onboarding agreements from a set of nodes of a MaaS network (e.g., the MaaS network **100**). The set of nodes may include at least one of a first set of mobility provider nodes (e.g., the first set of mobility provider nodes **104**) and a second set of MaaS player nodes (e.g., the second set of MaaS player nodes **106**). The operations may further include generation of a set of service and agreement templates based on the received set of onboarding agreements to create a MaaS node group (such as, the MaaS community **302**) corresponding to the set of nodes. The operations may further include transmission of the received set of onboarding agreements and the generated set of service and agreement templates to the created MaaS node group (such as, the MaaS community **302**). The operations may further include reception of an onboarding request from a first node of the MaaS network **100**. The operations may further include verification of a decentralized identity (DID) of the first node based on the received onboarding request. The operations may further include transmission of the first node, a set of parameters and a service agreement associated with the first node, based on the verified DID, transmitted set of onboarding agreements, and the set of service and agreement templates. The operations may further include transmission of the service agreement associated with the first node to the created MaaS node group (such as, the MaaS community **302**). The operations may further include generation of an API key of a data hub for the MaaS network **100**, based on the received service agreement. The first node may be onboarded to the created MaaS node group (such as, the MaaS community **302**) of the MaaS network **100** based on the generated API key and the transmitted service agreement associated with the first node.

Exemplary aspects of the disclosure may provide an electronic device (such as, an electronic device **102** of FIG. **1**) that includes circuitry (such as, the circuitry **1002**). The circuitry **1002** may be configured to receive the set of onboarding agreements from a set of nodes of a MaaS network (such as the MaaS network **100**). The set of nodes may include at least one of a first set of mobility provider nodes (such as, the first set of MP nodes **104**) and a second

set of MaaS player nodes (such as, the second set of MaaS player nodes **106**). The set of onboarding agreements may correspond to at least one of a master service agreement, a terms-of-service agreement, or a privacy policy agreement. In an example, the set of onboarding agreements may be received from the second set of MaaS player nodes **106**. The MaaS network **100** may further include at least one of an issuer node (such as the issuer node **112**) associated with the credential issuing authority, a set of holder nodes (such as the set of holder nodes **114**) corresponding to the set of nodes, a verifier node (such as the verifier node **116**) associated with a MaaS network (such as the MaaS network **100**) owner, and a distributed ledger database (such as, the first distributed ledger database **118** and the second distributed ledger database **120**).

Further, the circuitry **1002** may be configured to generate a set of service and agreement templates (such as the set of service and agreement templates) based on the received set of onboarding agreements to create a MaaS node group (such as, the MaaS community **302**) corresponding to the set of nodes. Further, the circuitry **1002** may be configured to transmit the received set of onboarding agreements and the generated set of service and agreement templates to the created MaaS node group (such as, the MaaS community **302**). In an embodiment, the MaaS community **302** may generate the MSA, the ToS, and the PP.

Further, the circuitry **1002** may be configured to receive an onboarding request from a first node (for example, the second set of MaaS player nodes **106**) of the MaaS network **100**. In an example, the onboarding request may be received from the set of users **216** through the second set of MaaS player nodes **106**. The circuitry **1002** may further verify a decentralized identity (DID) of the first node based on the received onboarding request.

Further, the circuitry **1002** may be configured to transmit, to the first node, a set of parameters (such as the set of parameters) and a service agreement (such as the service agreement) associated with the first node, based on the verified DID, the transmitted set of onboarding agreements, and the set of service and agreement templates.

Further, the circuitry **1002** may be configured to transmit the service agreement associated with the first node to the created MaaS node group (such as, the MaaS community **302**). The circuitry **1002** may be configured to generate an API key of a data hub for the MaaS network **100**, based on the received service agreement. The first node may be onboarded to the created MaaS node group (such as, the MaaS community **302**) of the MaaS network **100** based on the generated API key and the transmitted service agreement associated with the first node.

In an embodiment, each of the issuer node **112**, the set of holder nodes **114**, and the verifier node **116** may be associated with the second distributed ledger database **120** and may include a corresponding DID agent.

In an embodiment, the issuer node **112** may be configured to generate credentials associated with each holder node of the set of holder nodes **114**.

In an embodiment, the generated credentials may correspond to a set of attributes including at least one of an organization name, an organization ID, a name and electronic-mail ID of an administrator user, a role of a user of a node, or a user privilege level associated with a user of a node.

In an embodiment, the verifier node may be configured to receive a credential verification request associated with the first node, based on the received onboarding request of the first node. The verifier node may further query a holder node

of the set of holder nodes **114** for verified credentials of the first node, based on the received credential verification request. Thereafter, the verifier node **116** may be configured to receive, from the holder node, the verified credentials of the first node, based on the query. The DID of the first node may be verified based on the received verified credentials of the first node.

In an embodiment, the circuitry **1002** may be further configured to create a user profile and a role for the user profile, associated with the first node in the MaaS network **100**. The creation of the user profile and the role may be based on the verified DID of the first node and a scheduler (e.g., the scheduler **416**).

In an embodiment, the circuitry **1002** may further be configured to receive, from the first MaaS player node of the second set of MaaS player nodes **106**, the first master service agreement (MSA) and counterparty details associated with the first MaaS player node. The counterparty details may correspond to the first mobility provider node of the first set of mobility provider nodes **104**. The circuitry **1002** may be further configured to transmit, to the first mobility provider node, the first MSA, based on received the counterparty details. Thereafter, the circuitry **1002** may be configured to receive, from the first mobility provider node, an input indicative of an acceptance of the transmitted first MSA. The first MSA may be established between the first MaaS player node and the first mobility provider node, based on the received input indicative of the acceptance of the transmitted first MSA.

In an embodiment, the circuitry **1002** may further be configured to receive, from the first MaaS player node, the first service agreement associated with the first MaaS player node. The circuitry **1002** may further be configured to associate the received first service agreement with process service information and the first MSA, to determine an MSA bound service agreement. The circuitry **1002** may further be configured to transmit, to the first mobility provider node, the determined MSA-bound service agreement. The circuitry **1002** may further be configured to receive, from the first mobility provider node, an input indicative of an acceptance of the transmitted MSA-bound service agreement. The first service agreement may be established between the first MaaS player node and the first mobility provider node, based on the received input indicative of the acceptance of the transmitted MSA-bound service agreement.

In an embodiment, the circuitry **1002** may further be configured to receive, from the first MaaS player node, the first terms-of-service (ToS) associated with the first MaaS player node. Further, the circuitry **1002** may be configured to associate the received first ToS with the first service agreement to determine the service agreement bound-ToS. Further, the circuitry **1002** may be configured to transmit, to the first mobility provider node, the determined service agreement bound-ToS. Thereafter, the circuitry **1002** may be configured to receive, from the first mobility provider node, an input indicative of an acceptance of the transmitted service agreement bound-ToS. The first ToS may be established between the first MaaS player node and the first mobility provider node, based on the received input indicative of the acceptance of the transmitted service agreement bound-ToS.

The present disclosure may be realized in hardware, or a combination of hardware and software. The present disclosure may be realized in a centralized fashion, in at least one computer system, or in a distributed fashion, where different elements may be spread across several interconnected com-

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puter systems. A computer system or other apparatus adapted to carry out the methods described herein may be suited. A combination of hardware and software may be a general-purpose computer system with a computer program that, when loaded and executed, may control the computer system such that it carries out the methods described herein. The present disclosure may be realized in hardware that includes a portion of an integrated circuit that also performs other functions.

The present disclosure may also be embedded in a computer program product, which includes all the features that enable the implementation of the methods described herein, and which when loaded in a computer system is able to carry out these methods. Computer program, in the present context, means any expression, in any language, code or notation, of a set of instructions intended to cause a system with information processing capability to perform a particular function either directly, or after either or both of the following: a) conversion to another language, code or notation; b) reproduction in a different material form.

While the present disclosure is described with reference to certain embodiments, it will be understood by those skilled in the art that various changes may be made, and equivalents may be substituted without departure from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departure from its scope. Therefore, it is intended that the present disclosure is not limited to the particular embodiment disclosed, but that the present disclosure will include all embodiments that fall within the scope of the appended claims.

What is claimed is:

1. An electronic device, comprising:
circuitry configured to:

receive a set of onboarding agreements from a set of nodes of a mobility-as-a-service (MaaS) network, wherein the set of nodes includes at least one of a first set of mobility provider nodes and a second set of MaaS player nodes;

generate a set of service and agreement templates based on the received set of onboarding agreements to create a MaaS node group corresponding to the set of nodes;

transmit the received set of onboarding agreements and the generated set of service and agreement templates to the created MaaS node group;

receive an onboarding request from a first node of the MaaS network;

verify a decentralized identity (DID) of the first node based on the received onboarding request;

transmit, to the first node, a set of parameters and a service agreement associated with the first node, based on the verified DID, transmitted set of onboarding agreements, and the set of service and agreement templates;

transmit the service agreement associated with the first node to the created MaaS node group; and

generate an Application Programming Interface (API) key of a data hub for the MaaS network, based on the DID of the first node and the transmitted service agreement, wherein

the first node is onboarded to the created MaaS node group of the MaaS network based on the generated API key and the transmitted service agreement associated with the first node.

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2. The electronic device according to claim 1, wherein the set of onboarding agreements corresponds to at least one of a master service agreement, a terms-of-service agreement, or a privacy policy agreement.

3. The electronic device according to claim 1, wherein the MaaS network includes at least one of:

an issuer node associated with a credential issuing authority,

a set of holder nodes corresponding to the set of nodes, a verifier node associated with a MaaS network owner, and

a distributed ledger database.

4. The electronic device according to claim 3, wherein each of the issuer node, the set of holder nodes, and the verifier node, is associated with the distributed ledger database, and

each of the issuer node, the set of holder nodes, and the verifier node includes a corresponding DID agent.

5. The electronic device according to claim 3, wherein the issuer node generates credentials associated with each holder node of the set of holder nodes.

6. The electronic device according to claim 5, wherein the generated credentials correspond to a set of attributes including at least one of an organization name, an organization ID, a name and electronic-mail ID of an administrator user, a role of a user of a node, or a user privilege level associated with the user of the node.

7. The electronic device according to claim 3, wherein the verifier node:

receives a credential verification request associated with the first node, based on the received onboarding request of the first node;

queries a holder node of the set of holder nodes for verified credentials of the first node, based on the received credential verification request; and

receives, from the holder node, the verified credentials of the first node, based on the query, wherein the DID of the first node is verified based on the received verified credentials of the first node.

8. The electronic device according to claim 1, wherein the circuitry is further configured to create a user and a role for the user, associated with the first node in the MaaS network, and

the creation of the user and the role is based on the verified DID of the first node and a scheduler.

9. The electronic device according to claim 1, wherein the circuitry is further configured to:

receive, from a first MaaS player node of the second set of MaaS player nodes, a first master service agreement (MSA) and counterparty details associated with the first MaaS player node, wherein

the counterparty details correspond to a first mobility provider node of the first set of mobility provider nodes;

transmit, to the first mobility provider node, the first MSA, based on received the counterparty details; and receive, from the first mobility provider node, an input indicative of an acceptance of the transmitted first MSA, wherein

the first MSA is established between the first MaaS player node and the first mobility provider node, based on the received input indicative of the acceptance of the transmitted first MSA.

10. The electronic device according to claim 9, wherein the circuitry is further configured to:

receive, from the first MaaS player node, a first service agreement associated with the first MaaS player node;

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associate the received first service agreement with process service information and the first MSA, to determine an MSA bound service agreement;

transmit, to the first mobility provider node, the determined MSA bound service agreement; and

receive, from the first mobility provider node, an input indicative of an acceptance of the transmitted MSA bound service agreement, wherein

the first service agreement is established between the first MaaS player node and the first mobility provider node, based on the received input indicative of the acceptance of the transmitted MSA bound service agreement.

11. The electronic device according to claim 10, wherein the circuitry is further configured to:

receive, from the first MaaS player node, a first terms-of-service (ToS) associated with the first MaaS player node;

associate the received first ToS with the first service agreement to determine a service agreement bound-ToS;

transmit, to the first mobility provider node, the determined service agreement bound-ToS; and

receive, from the first mobility provider node, an input indicative of an acceptance of the transmitted service agreement bound-ToS, wherein

the first ToS is established between the first MaaS player node and the first mobility provider node, based on the received input indicative of the acceptance of the transmitted service agreement bound-ToS.

12. The electronic device according to claim 11, wherein the circuitry is further configured to:

receive, from the first MaaS player node, a second ToS associated with the first MaaS player node, wherein the second ToS corresponds to a new version of the established first ToS;

transmit, to the first mobility provider node, the second ToS;

receive, from the first mobility provider node, an input indicative of an acceptance of the second ToS; and

disable the first ToS established between the first MaaS player node and the first mobility provider node, based on the received input indicative of the acceptance of the second ToS, wherein

the second ToS is established between the first MaaS player node and the first mobility provider node, based on the disablement of the first ToS.

13. A method, comprising:

in an electronic device:

receiving a set of onboarding agreements from a set of nodes of a mobility-as-a-service (MaaS) network, wherein

the set of nodes includes at least one of a first set of mobility provider nodes and a second set of MaaS player nodes;

generating a set of service and agreement templates based on the received set of onboarding agreements to create a MaaS node group corresponding to the set of nodes;

transmitting the received set of onboarding agreements and the generated set of service and agreement templates to the created MaaS node group;

receiving an onboarding request from a first node of the MaaS network;

verifying a decentralized identity (DID) of the first node based on the received onboarding request;

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transmitting, to the first node, a set of parameters and a service agreement associated with the first node, based on the verified DID, transmitted set of onboarding agreements, and the set of service and agreement templates;

transmitting the service agreement associated with the first node to the created MaaS node group; and

generating an Application Programming Interface (API) key of a data hub for the MaaS network, based on the DID of the first node and the transmitted service agreement, wherein

the first node is onboarded to the created MaaS node group of the MaaS network based on the generated API key and the transmitted service agreement associated with the first node.

14. The method according to claim 13, wherein the set of onboarding agreements corresponds to at least one of a master service agreement, a terms-of-service agreement, or a privacy policy agreement.

15. The method according to claim 13, wherein the MaaS network includes at least one of:

an issuer node associated with a credential issuing authority,

a set of holder nodes corresponding to the set of nodes,

a verifier node associated with a MaaS network owner, and

a distributed ledger database.

16. The method according to claim 15, wherein each of the issuer node, the set of holder nodes, and the verifier node is associated with the distributed ledger database and

each of the issuer node, the set of holder nodes, and the verifier node includes a corresponding DID agent.

17. The method according to claim 15, wherein the issuer node generates credentials associated with each holder node of the set of holder nodes, and the generated credentials correspond to a set of attributes including at least one of an organization name, an organization ID, a name and electronic-mail ID of an administrator user, a role of a user of a node, or a user privilege level associated with the user of the node.

18. The method according to claim 15, wherein the verifier node:

receives a credential verification request associated with the first node, based on the received onboarding request of the first node;

queries a holder node of the set of holder nodes for verified credentials of the first node, based on the received credential verification request; and

receives, from the holder node, the verified credentials of the first node, based on the query, wherein

the DID of the first node is verified based on the received verified credentials of the first node.

19. The method according to claim 13, further comprising creating a user and a role for the user, associated with the first node in the MaaS network, wherein

the creation of the user and the role is based on the verified DID of the first node and a scheduler.

20. A non-transitory computer-readable medium having stored thereon, computer-executable instructions that when executed by an electronic device, causes the electronic device to execute operations, the operations comprising:

receiving a set of onboarding agreements from a set of nodes of a mobility-as-a-service (MaaS) network, wherein

the set of nodes includes at least one of a first set of mobility provider nodes and a second set of MaaS player nodes;
generating a set of service and agreement templates based on the received set of onboarding agreements to create 5
a MaaS node group corresponding to the set of nodes;
transmitting the received set of onboarding agreements and the generated set of service and agreement templates to the created MaaS node group;
receiving an onboarding request from a first node of the 10
MaaS network;
verifying a decentralized identity (DID) of the first node based on the received onboarding request;
transmitting, to the first node, a set of parameters and a service agreement associated with the first node, based 15
on the verified DID, transmitted set of onboarding agreements, and the set of service and agreement templates;
transmitting the service agreement associated with the first node to the created MaaS node group; and 20
generating an Application Programming Interface (API) key of a data hub for the MaaS network, based on the DID of the first node and the transmitted service agreement, wherein
the first node is onboarded to the created MaaS node 25
group of the MaaS network based on the generated API key and the transmitted service agreement associated with the first node.

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